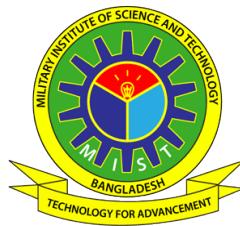


VISUALIZING BENGALI SPEECH WITH LIPNET AND MEDIAPIPE

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MD. TASHRIF AHSAN**

B.Sc. ENGINEERING THESIS



**DEPARTMENT OF ELECTRICAL, ELECTRONIC AND
COMMUNICATION ENGINEERING
MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY
DHAKA, BANGLADESH**

MARCH 2024

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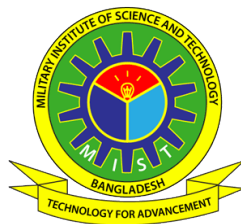
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A Thesis Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Electrical, Electronic and
Communication Engineering



DEPARTMENT OF ELECTRICAL, ELECTRONIC AND
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VISUALIZING BENGALI SPEECH WITH LIPNET AND MEDIAPIPE

DECLARATION

We hereby declare that the study reported in this thesis entitled as above is our own original work and has not been submitted before anywhere for any degree or other purposes. Further, we certify that the intellectual content of this thesis is the product of our own work and that all the assistance received in preparing this thesis and sources have been acknowledged and cited in the reference section.

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ABSTRACT

Visualizing Bengali speech with LipNet and Mediapipe

Visual speech recognition (VSR), or lipreading, is an essential part of multimodal communication that involves analysing lip and facial gestures and forms to decode speech. Within the expansive realm of artificial intelligence and human-computer interaction, this thesis targets pioneering a deep learning model designed for lipreading in the Bengali language.

This thesis intends to develop an deep neural network model for advanced machine lipreading for Bengali language by using the largest Bengali lip-reading dataset. For that, a hybrid deep learning architecture is used, combining 3D-CNN for spatial feature extraction and bi-directional LSTM for temporal decoding. The proposed model held an accuracy rate of 84.61% after thorough evaluation, highlighting its effectiveness. The key findings involve improving the underlying model for the Bengali language. An efficient pre-processing method has been developed to improve facial landmark detection in videos captured on mobile devices, reducing the time and cost required for processing image frames. The fully trained model's responsiveness allows for easy deployment on embedded systems, making visual or audiovisual communication in Bengali more accessible. This work thoroughly analyses previous lipreading models, providing significant information for future researchers. In summation, this thesis not only advances the frontiers of Bengali lipreading through novel methodologies but also lays a robust foundation for future research endeavours in this specialised area, contributing to the broader discourse on artificial intelligence and human-computer interaction.

সারসংক্ষেপ

Visualizing Bengali Speech With LipNet and Mediapipe

ভিজুয়াল স্পিচ রিকগনিশন বা লিপরিডিং, মাল্টিমোডাল যোগাযোগের একটি অপরিহার্য অংশ যা ভাষার অর্থোদ্রার করার জন্য ঠোঁট এবং মুখের অঙ্গভঙ্গি এবং ফর্মগুলি বিশ্লেষণ করে। কৃত্রিম বুদ্ধিমত্তা এবং মানব-কম্পিউটার মিথষ্ক্রিয়ার (Human Computer Interfacing) বিস্তৃত ক্ষেত্রের মধ্যে, এই থিসিসটি বাংলা ভাষায় লিপরিডিংয়ের জন্য একটি ডিপ লার্নিং মডেল বিকাশের লক্ষ্য রাখে।

এই থিসিসের উদ্দেশ্য হল সবচেয়ে বড় বাংলা লিপ-রিডিং ডেটাসেট ব্যবহার করে বাংলা ভাষার জন্য উন্নত মেশিন লিপরিডিংয়ের একটি ডিপ নিউরাল নেটওয়ার্ক মডেল তৈরি করা। এর জন্য, একটি হাইব্রিড ডিপ লার্নিং আর্কিটেকচার ব্যবহার করা হয়, যা স্থানিক বৈশিষ্ট্য (Spatial feature) উত্তোলনের জন্য থ্রিডি সিএনএন এবং সাময়িক ডিকোডিংয়ের (Temporal decoding) জন্য দ্বি-দিকনির্দেশক এলএসটিএমকে (Bidirectional LSTM) একত্রিত করে। প্রস্তাবিত মডেলটি পুঙ্খানুপুঙ্খ মূল্যায়নের পরে এর কার্যকারিতা তুলে ৮৪.৬১% পাওয়া যায় যা এর নির্ভুলতার হার ধারণ করে। মূল অনুসন্ধানগুলি বাংলা ভাষার অন্তর্নিহিত মডেলের উন্নতির সাথে জড়িত। মোবাইল ডিভাইসে ধারণ করা ভিডিওগুলিতে মুখের ল্যান্ডমার্ক সনাক্তকরণের উন্নতির জন্য একটি দক্ষ প্রাক-প্রক্রিয়াকরণ পদ্ধতি তৈরি করা হয়েছে, যা চিত্র ফ্রেমগুলি প্রক্রিয়াকরণের জন্য প্রয়োজনীয় সময় এবং ব্যয় হ্রাস করে। সম্পূর্ণরূপে প্রশিক্ষিত মডেলের প্রতিক্রিয়াশীলতা এমবেডেড সিস্টেমে সহজ স্থাপনার অনুমতি দেয়, যা বাংলায় ভিজুয়াল বা অডিওভিজুয়াল যোগাযোগকে আরও অ্যাক্সেসযোগ্য করে তোলে। পূর্ববর্তী লিপরিডিং মডেলগুলি পুঙ্খানুপুঙ্খভাবে বিশ্লেষণ করায় এই কাজটি ভবিষ্যতের গবেষকদের জন্য উল্লেখযোগ্য তথ্য সরবরাহ করে। সংক্ষেপে, এই থিসিসটি কেবল অভিনব পদ্ধতির মাধ্যমে বাংলা লিপরিডিংয়ের সীমানা এগিয়ে নিয়ে যায় না, এই বিশেষ ক্ষেত্রে ভবিষ্যতের গবেষণা প্রচেষ্টার জন্য একটি শক্তিশালী ভিত্তি স্থাপন করে, যা কৃত্রিম বুদ্ধিমত্তা এবং মানব-কম্পিউটার মিথষ্ক্রিয়া সম্পর্কিত বিস্তৃত আলোচনায় অবদান রাখে।

LIST OF NOTATIONS

X	The input feature map
W	The set of learnable filter
b	The bias term
Z	The output feature map
$*$	The convolution operation
$Z_{i,j}$	The value at position (i, j) in the output feature map Z
$W_{m,n}$	The weight at position (m, n) in the filter W
$X_{i+m,j+n}$	The input values covered by the filter at position (m, n) when centered at position (i, j) in the input feature map X
σ	The activation function applied element-wise after the convolution operation
O	The output size
N	The input size
F	The filter size
P	Padding
S	The stride
h_t	The hidden state vector at time step t
x_t	The input vector at time step t
W_{ih}	The weight matrix connecting input to hidden state
W_{hh}	The weight matrix connecting hidden state to hidden state
b_h	The bias vector for the hidden state
h_{t-1}	The hidden state vector at the previous time step
y_t	The output vector at time step t
W_{hy}	The weight matrix connecting hidden state to output
b_y	The bias vector for the output
$\mathbb{R}^n$	denotes the n -dimensional real vector space
W_z, W_r, W_h	Weight matrices of the update gate, reset gate and candidate activation vector respectively
e_{ij}	The attention score between encoder hidden state h_i and decoder hidden state s_j
W_a, v_a, b_a	Learnable parameters
$\tanh$	The hyperbolic tangent function

$h_i; s_j$	The concatenation of encoder hidden state h_i and decoder hidden state s_j
T	The length of the input sequence
x_t^{fused}	The fused feature vector
d	The dimensionality of the fused representation
π	The probability of a particular alignment
$y_{\pi_t}^t$	The probability of emitting the symbol π_t at time step t , as predicted by the neural network
Y	The total probability
$B(Y)$	The set of all possible alignments that can be collapsed into the target sequence Y
β_1	Exponential decay rate for the first-moment estimates (typical value: 0.9)
β_2	Exponential decay rate for the second-moment estimates (typical value: 0.999)
ε	Small constant for numerical stability (typical value: 10^{-8})
g_t	the gradient of the loss function with respect to the model's parameters
α	Learning rate

LIST OF ABBREVIATIONS

VSR	Visual Speech Recognition
ASR	Automated Speech Recognition
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
AAM	Active Appearance Model
DL	Deep Learning
HMM	Hidden Markov Model
TCN	Temporal Convolution Network
ASM	Active Shape Models
PCA	Principal Component Analysis
GRU	Gated Recurrent Unit
WLAS	Watch, Listen, Attend and Spell
MSE	Mean Square Error
MAE	Mean Absolute Error
BCE	Binary Cross-Entropy
CCE	Categorical Cross-Entropy
SVM	Support Vector Machine
CTC	Connectionist Temporal Classification
SGD	Stochastic Gradient Descent
ROI	Region of Interest
GNN	Graph Neural Network
RBM	Restricted Boltzmann Machines
DCT	Discrete Cosine Transform
DWT	Discrete Wavelet Transform
WER	Word Error Rate
CER	Character Error Rate
Adam	Adaptive Moment Estimation

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CHAPTER 1

INTRODUCTION

1.1 Background

Lip-reading, also known as Visual Speech Recognition (VSR), is the process of understanding speech by observing the movements of a speaker's lips, facial expressions, and other visual cues. Lip-reading models are computational systems designed to automatically interpret these visual cues to recognise spoken words or phrases. The primary motivation behind the development of lip-reading models is to provide communication assistance to individuals who are deaf or hard of hearing, as well as to enhance human-computer interaction in noisy environments or situations where audio cues are insufficient or unavailable. Additionally, lip-reading models have applications in surveillance, forensic analysis, and human-computer interfaces for assistive technologies.

The evolution of lip-reading models can be traced back to the early development of automated speech recognition (ASR) systems in the 1950s and 1960s. Initially, ASR systems focused primarily on analysing audio signals to transcribe spoken language. However, researchers soon realised the potential of integrating visual information, particularly lip movements, to improve speech recognition accuracy, especially in noisy environments. Early lip-reading models relied heavily on handcrafted features and rule-based approaches to analyse lip movements and extract phonetic information. These models often struggled with variations in lighting conditions, speaker articulation, and occlusions, limiting their effectiveness in real-world scenarios.

With advancements in computer vision and machine learning techniques, particularly the rise of deep learning (DL), the landscape of lip-reading models underwent a significant transformation. Convolutional Neural Networks (CNNs) emerged as the cornerstone of modern lip-reading models, enabling end-to-end learning of features directly from raw visual data. The

development of large-scale datasets specifically curated for lip-reading, such as the GRID corpus and the LRW dataset, played a crucial role in advancing the state-of-the-art in lip-reading models. These datasets provided researchers with ample training data to train deep neural networks effectively, leading to substantial improvements in lip-reading accuracy and robustness.

Over time, lip-reading models evolved from simple frame-based classifiers to sophisticated spatiotemporal architectures capable of capturing both spatial and temporal dynamics of lip movements. Hybrid models combining CNNs with recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM) networks, became prevalent, enabling lip-reading systems to model long-range dependencies in sequential visual data. Furthermore, attention mechanisms were introduced to lip-reading models to focus on relevant regions of the lip movements dynamically, enhancing the interpretability and performance of the models. These attention-based architectures, combined with multi modal fusion techniques incorporating both visual and audio modalities, achieved state-of-the-art results in lip-reading benchmarks.

Lip-reading models have undergone a remarkable evolution from early rule-based systems to DL-based architectures capable of robustly recognising spoken words from visual cues. The integration of advanced machine learning techniques, large-scale datasets, and multimodal fusion strategies has propelled the field of lip-reading towards more accurate, reliable, and accessible communication technologies

1.2 Literature Review

The literature review on lipreading models reveals a rich landscape of research efforts aimed at developing effective systems for various speech recognition tasks. The AVLetters database has been pivotal in advancing alphabet recognition, with notable contributions from Zhao et al. [49] and Pei et al. [137]. Zhao et al. employed LBP-TOP for feature extraction coupled with Support Vector Machine (SVM) classification, achieving a word accuracy rate (WAR) of 62.80 %. On the other hand, Pei et al. recorded the highest WAR of 69.60 % using an

RFMA-based lip-reading system.

For digit recognition, the CUAVE database has been extensively utilised, with Papandreou et al. [139] and Rahmani and Almasganj [140] among the notable contributors. Papandreou et al. utilised an Active Appearance Model (AAM) for feature extraction and Hidden Markov Model (HMM) for classification, achieving a word recognition rate of 83.00%. Meanwhile, Rahmani and Almasganj achieved a WAR of 63.40% by employing deep bottleneck features with a GMM-HMM language model.

The GRID database, focusing on phrase prediction, has seen significant exploration, particularly by Wand et al. [112] and Assael et al. [91]. Wand et al. experimented with various feature extraction techniques, highlighting the effectiveness of a feedforward network coupled with a Long LSTM backend. Assael et al., along with Xu et al. and Margam et al., achieved impressive word accuracies ranging from 95.20% to 98.70% using spatiotemporal convolutional networks and Bidirectional RNNs .

In the world of multi-view databases like OuluVS2, Lee et al. [85] and Wu et al. [141] have made significant strides. Lee et al. combined Discrete Cosine Transform (DCT) and Principal Component Analysis (PCA) features with HMM classification, achieving a word accuracy rate of 63.00%. Wu et al. achieved a high classification accuracy of 87.55% by combining SDF and STLP features with SVM classification.

LRW dataset, known for its challenging nature, has been extensively explored for word classification. Chung and Zisserman [40] and Torfi et al. [92] demonstrated notable achievements, with word accuracy rates ranging from 61.10% to 98.50%. Recent advancements by Zhang et al., Xiao et al., Luo et al., and Zhao et al. have further pushed the boundaries, with state-of-the-art WARs ranging from 83.50% to 85.20% by employing 3D CNNs, ResNet, and Bidirectional Gated Recurrent Units (GRUs).

The literature also highlights the emergence of DL methods, particularly Transformers and

Temporal Convolutional Networks (TCNs), for word and sentence prediction tasks. These models offer advantages over traditional methods, especially in handling parallel computation and learning long-term dependencies . While traditional methods still outperform DL approaches in recognising shorter speech segments due to the need for large training samples, the trend is shifting towards leveraging DL for sentence prediction tasks.

1.3 Research Motivation

The motivation behind this thesis on Bangla lipreading extends beyond addressing the needs of the hearing-impaired community to exploring the diverse applications of lipreading technology across various domains. While the primary goal is to enhance communication accessibility for individuals with hearing impairments in the Bangla-speaking region, the potential applications of a robust Bangla lipreading model are manifold and far-reaching.

One significant application of lipreading technology is in enhancing security measures, particularly in high-risk environments such as banks and financial institutions. Integrating lipreading capabilities into existing security systems can bolster identity verification processes and mitigate security breaches by adding an additional layer of authentication. Furthermore, the ability to detect suspicious behaviour or unauthorised individuals through lipreading analysis can enhance overall security protocols.

Moreover, the medical field stands to benefit significantly from advancements in lipreading technology. For instance, in the detection and diagnosis of neurological disorders such as Parkinson's disease and stroke, subtle changes in speech and facial movements captured through lipreading can serve as valuable diagnostic indicators. Additionally, for cancer patients undergoing treatment, monitoring speech patterns and facial expressions through lipreading can aid in assessing pain levels and emotional well-being, enabling healthcare providers to deliver more personalised care.

Beyond security and healthcare, lipreading technology holds promise for various social appli-

cations. For example, in educational settings, integrating lipreading capabilities into learning platforms can facilitate language acquisition for individuals with hearing impairments, fostering greater inclusion in mainstream educational environments. Furthermore, in the context of human-computer interaction, lipreading technology can enable more intuitive and accessible interfaces, enhancing the user experience for individuals with diverse communication needs.

By exploring these diverse applications, this thesis aims to underscore the versatility and potential impact of Bangla lipreading technology across multiple domains. Through the development of a language-specific lipreading model for the Bangla-speaking community, this research seeks to pave the way for innovative solutions that not only address communication barriers but also contribute to advancements in security, healthcare, education, and human-computer interaction.

1.4 Research Objectives

Recognising the scarcity of Bengali lipreading resources, this thesis tries to enrich the field of Bengali machine lipreading. With the aim of breaking the language barrier, the specific objectives of this work are outlined as follows:

- a. To develop an optimal machine lipreading deep neural network model for the Bengali language.
- b. To train and evaluate the performance of the model utilising the largest available Bengali lip-reading dataset. ‘LipBengal’.
- c. To validate the accuracy of the model using prominent datasets, resulting in the creation of a pre-trained model that can be easily used on different embedded devices.

The expected outcome of this work is, therefore, a new dimension for Bengali VSR system providing a foundation to empower and assist new researchers in getting a head start in this field.

1.5 Layout of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 explains the basic terminologies required in order to understand DL based lip-reading. The chapter also portraits a historical analysis and change in lip-reading trend throughout the year.

Chapter 3 introduces to generalized process of DL based lip-reading and depicts the evolution of the techniques. The chapter also guides in steps towards the methods used in making our lip-reading model and explains it.

Chapter 4 presents the multi-angle assessment of the model and the experimental setup used for the assesment. Finally, analysis, comparative investigation of our work and some common performance metrices are shown.

Chapter 5 presents the conclusion of the thesis with a summary of the key findings. The chapter ends with instructions and scope for future work.

CHAPTER 2

RELATED WORKS

2.1 Introduction

Lip reading involves interpreting speech by observing the lip movements, and it has attracted significant interest in computer vision, machine learning, and assistive technology. Researchers have investigated several methods and strategies to create precise and resilient lip-reading systems that can comprehend spoken language relying just on visual information. This section offers an overview of previous studies in the field, tracking the development of lip reading research, emphasising significant approaches, and examining significant findings.

2.2 Early Approaches to Lip-Reading

Early lip-reading models represented the early phase of research in the field. They depended on manually designed features and algorithmic rules to analyse lip movements and extract phonetic data. The pioneering efforts laid the foundation for future progress but also exposed inherent challenges and limitations.

Petridis *et al.* [1] introduced a lip-reading system that focuses on Active Appearance Models (AAMs) and Active Shape Models (ASMs) for feature extraction. AAMs and ASMs are adaptive models that can depict variations in both shape and texture within images, enabling the portrayal of intricate lip movements. By aligning a set of template shapes to the lip region in video frames, these models attempted to encode the spatial dynamics of lip articulation. While these early methods showed promise in controlled environments, they struggled miserably in the face of real-world variability.

One of the biggest problems with early lip-reading models was that they weren't designed to be able to handle changes in the environment, like lighting. Variations in lighting can result in notable changes in the way lip areas look, which may result in problems in feature identifi-

cation and classification. Moreover, the variation in pronunciation among speakers presented a significant challenge due to the unique lip movements and speech patterns displayed by different individuals. This variability required the creation of models that can generalise across speakers and adapt to various articulatory styles.

Partial and complete occlusions posed significant difficulties for early lip-reading systems. Occlusions occur when objects block the view of the lips, hiding important visual cues needed for precise speech recognition. The occluding elements may include facial hair, accessories, hands, or nearby objects. To overcome occlusions, improved feature representations and complex algorithms must be used to identify important information among visual obstructions.

The absence of sufficient annotated datasets presented a major obstacle to initial lip-reading studies. Limited access to various and accurate datasets impeded researchers' capacity to efficiently train and assess their models. Lacking adequate data made it difficult to create models that could effectively apply to different speaking styles, languages, and environmental conditions.

Despite these challenges, the pioneering efforts of researchers in the early stages of lip-reading laid the foundation for subsequent advancements in the field. The early attempts highlighted the significance of strong feature representations, flexible algorithms, and extensive datasets in achieving precise and dependable lip-reading results.

2.3 Transition to Deep Learning (DL)

The advancement to DL was an important turning point in the development of lip-reading technology, facilitating the automatic extraction of hierarchical features from raw visual data. Convolutional Neural Networks (CNNs) are a prominent DL architecture known for effectively identifying spatial patterns in image frames, leading to major advancements in the field of lip-reading.

Noda *et al.* [2] were among the pioneers in utilising CNNs for lip-reading tasks. They demon-

strated that CNNs are more effective than traditional methods like Principal Component Analysis (PCA) in learning distinctive features from lip images. CNNs excel at automatically acquiring hierarchical representations by applying convolutional filters to input images. CNNs can effectively capture complex patterns and relationships between spaces in data by combining convolutional, pooling, and activation functions in multiple layers, which is why they are ideal for lip-reading tasks.

The shift to DL enabled end-to-end learning, allowing the entire lip-reading process to be trained as a whole, from raw input images to final classification results. This comprehensive method removed the necessity for manual feature engineering and allowed the model to autonomously acquire applicable representations from the data.

2.3.1 CNNs

CNNs [3] are DL models built for processing grid-like information, such as images. The structure includes several layers of convolutional filters, non-linear activation functions, and pooling layers. Mathematically, the forward pass of a convolutional layer in a CNN can be represented as follows: Let X be the input feature map, W be the set of learnable filters (also called kernels), b be the bias term, and Z be the output feature map. The convolution operation $*$ is applied between the input feature map X and the filters W as follows:

$$Z_{i,j} = \sigma \left(\sum_m \sum_n W_{m,n} \cdot X_{i+m,j+n} + b \right)$$

Where: $Z_{i,j}$ is the value at position (i, j) in the output feature map Z . $W_{m,n}$ is the weight at position (m, n) in the filter W . $X_{i+m,j+n}$ represents the input values covered by the filter at position (m, n) when centered at position (i, j) in the input feature map X . σ is the activation function applied element-wise after the convolution operation. b is the bias term added to each convolutional kernel operation. This equation computes the output feature map Z by convolving the input feature map X with the filters W , applying the activation function σ , and adding the bias term b .

Let's denote the input image as X , the weights of the convolutional layer as W , the bias as b , the output of the convolutional layer as Z , the activation function as σ , the pooling operation as P , the weights of the fully connected layer as W_f , the bias of the fully connected layer as b_f , and the final output as $\hat{y}$. The output of a convolutional layer is determined using the convolution operation, which is followed by an activation function like ReLU (Rectified Linear Unit) [4], and if necessary pooling to reduce spatial dimensions.

The equation for a convolutional layer can be represented as:

$$O = \frac{N - F + 2P}{S} + 1$$

where O is the output size, N is the input size, F is the filter size, P is the padding, and S is the stride.

2.3.2 Recurrent Neural Network (RNN)

RNNs [5] are designed to handle sequential data by maintaining a hidden state that captures information about previous time steps. This makes them suitable for modelling temporal dependencies in sequential data. The hidden state of an RNN is updated recursively based on the current input and the previous hidden state. The update equation for an RNN can be represented as: The hidden state h_t of an RNN at time step t is computed as follows:

$$h_t = \sigma(W_{ih}x_t + W_{hh}h_{t-1} + b_h)$$

where: $h_t \in \mathbb{R}^h$ is the hidden state vector at time step t . $x_t \in \mathbb{R}^d$ is the input vector at time step t . $W_{ih} \in \mathbb{R}^{h \times d}$ is the weight matrix connecting input to hidden state. $W_{hh} \in \mathbb{R}^{h \times h}$ is the weight matrix connecting hidden state to hidden state. $b_h \in \mathbb{R}^h$ is the bias vector for the hidden state. σ is the activation function, often the hyperbolic tangent function or the rectified linear unit (ReLU). h_{t-1} is the hidden state vector at the previous time step. The output y_t of the RNN at

time step t is computed as:

$$y_t = \sigma(W_{hy}h_t + b_y)$$

where: $y_t \in \mathbb{R}^m$ is the output vector at time step t . $W_{hy} \in \mathbb{R}^{m \times h}$ is the weight matrix connecting hidden state to output. $b_y \in \mathbb{R}^m$ is the bias vector for the output. The notation $\mathbb{R}^n$ denotes the n -dimensional real vector space.

2.3.2.1 Long Short-Term Memory(LSTM)

LSTMs [6] are a type of RNN architecture created to address the vanishing gradient problem encountered by conventional RNNs. They can efficiently capture extended relationships in sequential data. LSTMs retain a cell state to store information across extended sequences, making them well-suited for tasks that involve capturing temporal patterns.

2.3.2.2 Gated Recurrent Unit(GRU)

Gated Recurrent Unit (GRU) The GRUs [7] is a type of RNN architecture that solves the vanishing gradient problem encountered while using traditional RNNs. It is designed to capture dependencies in sequential data efficiently while reducing computational complexity. GRU does so through the use of gating mechanisms that control the flow of information within the network. The principal components of GRU are briefly stated above:

1. Update Gate (z_t): Decides how much of the past information to carry forward to the current time strap.
2. Reset Gate (r_t): Controls the amount of past data that will have to be forgotten. It contributes to solving the vanishing gradient problem.
3. Current Memory Content ($\tilde{h}_t$): Candidate activation function that might be added to the memory of GRU.
4. Hidden State (h_t): Represents the memory content at each time strap t . The update gate z_t , reset gate r_t , and candidate activation function $\tilde{h}_t$ are calculated as follows:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (2.1)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (2.2)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \quad (2.3)$$

where: sigma is the sigmoid activation function W_z, W_r , and W_h are weight matrices of the update gate, reset gate, and candidate activation vector, respectively.

b_z, b_r, b_h are bias vectors - $\odot$ denotes element-wise multiplication.

Finally, the hidden state h_t at timestamp t is updated by a linear interpolation between the previous hidden state h_{t-1} and the candidate activation vector $\tilde{h}_t$, controlled by the update gate z_t :

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (2.4)$$

This gives GRU the ability to selectively update its memory content based on the input at the current time step and the updated gates.

2.3.3 Attention Mechanisms:

Attention mechanisms have been incorporated into lip-reading models to selectively concentrate on important areas of lip movement, improving the model's capacity to extract significant information from visual input. One notable method in this regard is the Watch, Listen, Attend, and Spell (WLAS) system suggested by Chung *et al.* [8]. The lip-reading system integrates attention mechanisms to combine visual and audio information efficiently. This enables the model to focus on particular regions of the input data that are important for the current task.

Attention mechanisms involve assigning significance weights to various segments of the in-

put sequence, allowing the model to prioritise significant features and filter out irrelevant or noisy data. Attention mechanisms in lip-reading models help adjust focus dynamically during decoding to concentrate on the most informative areas of lip movements for precise transcription.

Given an encoder hidden state h_i and a decoder hidden state s_j at time step j , the attention score e_{ij} is computed as follows:

$$e_{ij} = \text{score}(h_i, s_j) = v_a^\top \tanh(W_a[h_i; s_j] + b_a)$$

where:

- e_{ij} is the attention score between encoder hidden state h_i and decoder hidden state s_j .
- W_a , v_a , and b_a are learnable parameters.
- $\tanh$ is the hyperbolic tangent function.
- $[h_i; s_j]$ denotes the concatenation of encoder hidden state h_i and decoder hidden state s_j .

The attention weights α_{ij} are computed using the softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^T \exp(e_{ik})}$$

where T is the length of the input sequence.

The context vector c_j , which represents the weighted sum of encoder hidden states, is computed as follows:

$$c_j = \sum_{i=1}^T \alpha_{ij} h_i$$

The attention mechanism allows the model to dynamically assign different weights to different parts of the input sequence, enabling it to focus on relevant information during decoding.

2.3.4 Multimodal Fusion:

Multimodal fusion techniques have emerged as a promising approach in the most recent advances of lip-reading models, aiming to leverage both visual and audio modalities to enhance performance. Stafylakis and Tzimiropoulos made a significant contribution by developing a visual speech recognition (VSR) system that incorporates word embeddings to condense important information obtained from lip movements. Their approach aims to enhance the robustness and accuracy of the lip-reading model by integrating information from various modalities to capture additional cues not fully represented by individual modalities alone.

The fundamental idea behind multimodal fusion is to combine information from different modalities in a meaningful way to enhance the overall representation of the input data. In the context of lip-reading, this typically involves integrating features extracted from visual (e.g., lip movements) and audio (e.g., speech signals) modalities to obtain a more comprehensive understanding of the spoken content.

Mathematically, multimodal fusion can be achieved through various techniques, such as concatenation, addition, or weighted combination of features from different modalities. One common approach is to concatenate the feature vectors from each modality and feed them into a fusion layer, which learns to combine the modalities' information effectively. This can be represented as follows:

Let x_t^{img} and x_t^{txt} denote the features from the image and text modalities, respectively, at time step t . The fused feature vector x_t^{fused} can be computed using a fusion function f as follows:

$$x_t^{\text{fused}} = f(x_t^{\text{img}}, x_t^{\text{txt}})$$

where $x_t^{\text{fused}} \in \mathbb{R}^d$ is the fused feature vector and d is the dimensionality of the fused representation.

There are various ways to perform multimodal fusion, including concatenation, element-wise

addition, element-wise multiplication, and more complex fusion functions such as neural networks or attention mechanisms.

Alternatively, fusion can be performed through element-wise addition or weighted combination of the feature vectors, followed by non-linear transformations to produce the fused representation. This allows the model to adaptively adjust the contribution of each modality based on their relevance to the task at hand.

2.3.5 Hybrid Architectures

Hybrid architectures that combine CNNs and RNNs are popular in lip-reading for their capacity to capture spatial and temporal aspects of lip movements. CNN-LSTM architectures merge the spatial feature extraction abilities of CNNs with the temporal modelling capabilities of LSTMs. CNNs extract spatial features from input frames, which are then inputted into LSTM layers to capture temporal dependencies across sequential frames.

The following Table- 2.1 shows the performance of different CNN-RNN based architectures applied to available datasets.

Table 2.1: RNN-based models applied on prominent datasets

Year	Reference	Classifier	Dataset	Class	Accuracy
2016	Chung and Zisserman [9]	LSTM	OuluVS2	Phrases	94.10%
2016	Petridis and Pantic [10]	LSTM	OuluVS2	Phrases	81.10%
2016	Garg et al [11]	LSTM	MIRACL-VC1	word+Phrases	76.00%
2017	Chung and Zisserman [12]	LSTM+attention	OuluVS2	Phrases	91.10%
2017	Petridis <i>et al.</i> [1]	Bidirectional-LSTM	OuluVS2	Phrases	94.70%
2017	Stafylakis and Tzimiropoulos [13]	Bi-directional LSTM	LRW	Words	88.08%
2018	Afourous <i>et al.</i> [14]	Bidirectional LSTM + language model	LRS2	Sentences	37.80%
2018	Petridis <i>et al.</i> [15]	Bidirectional-GRU	LRW	Words	82.00%
2018	Xu <i>et al.</i> [16]	Bidirectional-GRU+attention	GRID	Phrases	97.10%
2019	Weng and Kitami [17]	Bidirectional-LSTM	LRW-1000	Words	84.10%
2020	Zhang <i>et al.</i> [18]	Bidirectional-GRU	LRW	Words	84.13%
2020	Luo et al [19]	Bidirectional-GRU	LRW-1000	Words	38.57%
2020	Zhao et al [20]	Bidirectional-GRU	LRW-1000	Words	38.79%

2.3.6 Loss Functions and Optimizers

2.3.6.1 Loss Function

Loss functions are mathematical functions that quantify the differences between the predicted output of a model and the actual output. Loss functions are essential for backpropagation. The choice of loss function is very specific to specific problems. Some of the loss functions are:

- **Mean Square Error (MSE):** Commonly used for regression tasks (predicting continuous values), it measures the average squared difference between predicted and actual values. It is very easy to understand and implement but it is very sensitive to outliers.
- **Mean Absolute Error (MAE)** Another regression loss function, MAE calculates the average absolute difference between predictions and targets. Unlike MSE, MAE is less sensitive to outliers but it may not penalize large errors as heavily.
- **Binary Cross-Entropy (BCE)** Suitable for binary classification (two classes), it measures the difference between predicted probabilities and actual class labels. BCE handles probabilities well but it is not ideal for multi-class problems.
- **Categorical Cross-Entropy (CCE)** An extension of BCE for multi-class classification, it calculates the average cross-entropy across all classes. This error function handles multiple classes effectively, but similar to BCE, it may not be the best choice for imbalanced classes.
- **Hinge Loss** Often used in Support Vector Machines (SVMs) for classification, it aims to create a margin between correct and incorrect classifications. Hinge Loss function encourages large margins between classes, but it can be sensitive to noisy data.
- **Connectionist Temporal Classification (CTC Loss)** The function is designed for sequence-to-sequence problems with unknown or variable alignments between input and output sequences (e.g., speech recognition, handwriting recognition, and image se-

quence classification). The CTC Loss function handles unaligned data and considers all possible alignments, making it robust even when the exact timing of events within a sequence is uncertain but potentially computationally expensive. This can be more computationally intensive, especially for large output vocabularies.

2.3.6.2 Optimizers

Optimizers are algorithms or iterative procedures used in machine learning (ML) and DL to update the parameters (weights and biases) of a model in order to minimize a loss function and improve the model's performance on a given task. Optimizers can be assumed to be the engines that help DL models learn. They guide how the model should change its internal settings to get better at predicting the correct answer. Here are some popular optimizers:

- **Stochastic Gradient Descent (SGD)** A fundamental optimizer, SGD updates parameters based on the gradient of the loss function for a single data point in each iteration. It is simple and computationally inexpensive but can be slow to converge, especially with noisy data.
- **Momentum SGD** An extension of SGD that incorporates momentum to accumulate past gradients, smoothing the descent path and potentially accelerating convergence. It can overcome local minima more effectively than SGD but may not always converge or reach the optimal solution.
- **Adaptive Moment Estimation (Adam)** An advanced optimizer that combines ideas from SGD, momentum, and RMSprop (another optimizer) to adapt learning rates for each parameter based on their historical gradients. It often finds good solutions efficiently and is less prone to local minima. The only problem with it is that it can be computationally expensive for large models.

2.4 Future Research Scope

Despite the fact that lip-reading models have made significant progress, there are still various challenges that must be overcome to improve their effectiveness and practicality in real-world situations. One major challenge is maintaining resilience to changes in lighting conditions, which can greatly impact the quality of visual input recorded by lip-reading systems. Variations in lighting can cause discrepancies in lip appearance and texture, posing difficulties for models to precisely identify important characteristics for speech recognition. To tackle this issue, it is necessary to create methods that can efficiently manage different lighting conditions by either standardising input images or integrating strong feature extraction techniques that are not easily affected by changes in illumination.

Another important issue is the capacity of lip-reading models to accurately identify the speaker. Due to variations in lip shapes, sizes, and articulation patterns among individuals, models trained on data from certain speakers may have difficulty generalising to new speakers. This problem presents a major challenge in implementing lip-reading systems in practical scenarios where the speaker's identity can change. Future research may concentrate on creating speaker-invariant representations or utilising speaker adaptation techniques to enhance model generalisation across various speakers.

Occlusions pose a significant challenge to lip-reading by obstructing the view of the lips and reducing the quality of visual input. Obstructions can arise from factors like facial hair, facial accessories, or hand gestures, making it difficult to analyse lip movements accurately. To overcome this challenge, robust occlusion handling mechanisms need to be developed to accurately detect and compensate for occluded areas in input images. Methods like attention mechanisms or spatial-temporal reasoning can be investigated to concentrate on unobstructed areas of the lips and reduce the effects of occlusions on model performance.

One promising direction for future exploration is the use of novel architectures like 3D CNNs

to better capture spatiotemporal dynamics. 3D CNNs differ from 2D CNNs by incorporating temporal dependencies through the analysis of sequences of frames over time, in addition to spatial information within individual frames. 3D CNNs can enhance lip-reading models by utilising spatial and temporal data to better capture subtle lip movements and articulatory gestures, potentially improving accuracy.

Furthermore, creating extensive annotated datasets designed specifically for lip-reading tasks is essential for progressing in the field. Current datasets may be restricted in size, diversity, and annotation quality, which can impede the development and assessment of lip-reading models. Developing extensive datasets that include diverse speakers, speaking styles, and environmental conditions can help improve the reliability and applicability of lip-reading systems. Furthermore, integrating multi-view or multi-modal data sources like audio recordings or depth information can enhance datasets and facilitate the exploration of new research avenues in lip-reading.

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2.5 Chapter Summary

In summary, the field of lipreading has advanced in a galloping rate from the time of its beginning. Over 30 years, researchers have been contributing in this field with the generation of new lipreading datasets as well as by generation of DL based models.

With the advancement of the DL and machine learning field, researchers have been developing their architectures, incorporating single or even hybrid models. Among them 3DCNNs, RNNs, GRU, LSTM, Multimodal fusion or mixing of the upper mentioned methods have been used. Over the course of time the introduction of attention based models as well as transformer models have contributed a lot in the development of the DL architectures to perform lipreading efficiently.

CHAPTER 3

A CUSTOM DEEP LEARNING (DL) LIP-READING MODEL

3.1 Introduction

In the field of automated Visual Speech Recognition (VSR) systems, the process involves capturing video frames, isolating the Region of Interest (ROI), extracting spatio-temporal features, and classification. So, interpreting lip movements and converting them into text or other formats presents a complex spatio-temporal challenge. This multifaceted problem can be subdivided into several key components, including:

- **Data Collection** : The acquisition of a comprehensive dataset of videos featuring individual speakers.
- **Preprocessing**: The pre-processing of these videos involves tasks such as frame extraction and augmenting the data to enhance the diversity of lip movements captured.
- **Spatial Feature Extraction**: Using techniques such as convolutional neural networks (CNNs) to extract spatial features from individual frames. This step is done to capture the necessary spatial information from lip movements.
- **Temporal Feature Extraction**: Using recurrent neural networks (RNNs) or similar architectures to capture temporal dependencies between consecutive frames. This procedure makes it easier to figure out how changes in lip movements occur over time.
- **Combining Features and Classification**: Combining both spatial and temporal features extracted from the lip movement data. This integration is a must to decode speech from visual data.

Each of these components plays a vital role in the development of effective machine lipreading systems.

3.2 Evolution of Machine Lip-reading Models

The evolution of lip-reading started almost 30 years ago. But the development started to increase in a galloping rate when the advancement of DL field occurred. The development of lip-reading using traditional methods and DL based methods can be explained as follows:

3.2.1 Traditional Method

Before DL was introduced a pixel based or geometric based method or combination of both was used to isolate the ROI. Often a fixed slicing of images done manually to isolate the ROI. Traditional methods of VSR rely on extracting features from input video data using mathematical transformations like Active Appearance Models (AAMs) [21], Active Shape Models (ASMs) [21], Discrete Cosine Transform (DCT) [22], Principal Component Analysis (PCA) [23], and Discrete Wavelet Transform (DWT) [24]. These features are then classified using techniques such as Hidden Markov Models (HMM) [25] and Support Vector Machines (SVMs) [26]. However, traditional VSR approaches face limitations, including dependence on input video quality affected by factors like lighting and facial expressions, leading to variations in visual speech data that decrease the accuracy of recognition. Moreover, they struggle with handling pronunciation and dialect variations due to limited training data, and are computationally complex, slowing down processing and limiting real-time application, especially in noisy environments. Consequently, newer DL-based VSR approaches have emerged to overcome these limitations, promising more accurate and efficient speech recognition.

3.2.2 DL Based Methods

Lip-reading has significantly advanced with the development of DL. The use of advanced neural network models such as 3DCNN, RNNs, Long short Term Memory (LSTM), GRU, bi-directional LSTM, bi-directional GRU, and attention and transformer-based models significantly advanced the area of lip-reading. DL models can be used in the frontend, backend, or evaluation of a DL architecture. The frontend represents ROI detection or preprocessing, and the backend represents spatial and temporal feature extraction and classification.

Lip-reading models are highly language-dependent, though a transfer learning-based method was proposed by [27], which was tested on datasets of different languages and was less performing than average language-dependent models. So for training and testing, there was a need for a Bengali lipreading dataset, which was fulfilled by LipBengal [28]. The datasets listed below are publicly available to this date:

Table 3.1: Major publicly available lip-reading datasets

Dataset (Year)	Language	Data Type	Segment	Speakers	Classes	Utterances	Resolution	Pose
TULIPS [29] (1995)	English	Isolated	Digits	12	4	96	100x75	Frontal
AVLetters [30] (1998)	English	Isolated	Alphabet	10	26	780	376x288	Frontal
CAVSR1.0 [31] (2000)	Chinese	Isolated	Words	20	78	3120	352x228	Frontal
AVICAR [32] (2004)	Spanish	Continuous	Digits	86	26	59000	720x480	4 views
CUAVE [33] (2004)	English	Isolated	Digits	36	10	7000	720x480	90,0,90
GRID [34] (2006)	English	Isolated	Phrases	34	34000	34000	720x576	Frontal
CENSREC-AV-1 [35] (2010)	Japanese	Continuous	Digits	42	10	3234	720x480	Frontal
OuluVS [36] (2010)	English	Isolated	Phrases	53	10	1590	1920x1080	0,30,45,60,90
OuluVS2 [37] (2010)	English	Isolated	Sentences	53	540	2120	1920x1080	0,30,45,60,90
MIRACL-VC [38] (2011)	English	Isolated	Words	15	10	1500	640x480	Frontal
LRW [12] (2016)	English	Continuous	Words	10k+	500	400000	256x256	-30,30
LRS2 [8] (2017)	English	Isolated	Sentences	10k+	17428	118116	160x160	-30,30
LRS3 [39] (2018)	English	Isolated	Sentences	10k+	70000	165000	224x224	-90,90

3.3 General Steps for Applying Machine Lip-reading

There are several steps that are involved in machine lip reading including preprocessing, feature extraction, classification etc. They have been discussed in brief:

3.3.1 Preprocessing

In terms of VSR Preprocessing means extracting the ROI, which can be the face, mouth, or just the lip portion. Preprocessing in the field of VSR has evolved a lot. There are mainly four types of preprocessing techniques, according to Dupont and Luetttin [28]: geometric based, image-based, model-based, and motion based. However, the majority of the DL models can classify images so image based preprocessing is the most popular method. Most of the DL models used different approaches such as Dlib[40] a DL based face recognition technology to detect faces and i.bug [41] a facial landmark detector for detecting facial land marks and isolating the ROI. The face localization method using Dlib and facial landmarks detection using iBUG has been shown in figure- 3.1 and figure- 3.2

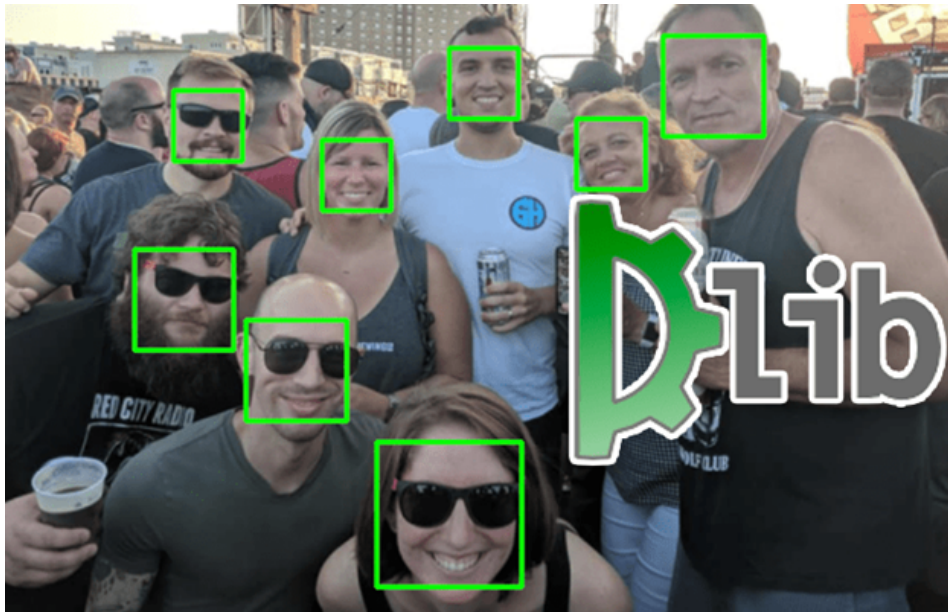


Figure 3.1: Detection of faces using Dlib



Figure 3.2: Identifying facial landmarks using iBUG

After isolating ROI several other preprocessing steps are involved including: grayscale conversion, z-score normalization and augmentation.

3.3.2 Spatial Feature Extraction

Spatial feature extraction aims to simplify image data by removing unnecessary spatial details, retaining only the essential features. This process reduces the complexity of the images, making them easier to analyze and work with. In DL models feature extraction are automated by using various processes such as:

- CNNs, including feedforward neural networks, 2D CNNs, 3D CNNs, and combinations thereof, serve as powerful tools for spatial feature extraction in image processing tasks.
- Attention Mechanisms stand out as effective spatial feature extractors due to their ability to selectively focus on specific regions within input images, enhancing model performance.
- Transformer-based models have emerged as promising alternatives, offering superior performance in spatial feature extraction tasks.

- Graph Neural Networks (GNNs) present another avenue for extracting spatial features, particularly advantageous for processing graph-structured data and capturing complex spatial relationships.
- Previously, techniques such as autoencoders and Restricted Boltzmann Machines (RBMs) have been utilized for feature extraction in lip-reading tasks, showcasing their potential in capturing meaningful spatial information from image data.

3.3.3 Temporal Feature Extraction

As video and sequential image frames are time series data so we need to somehow use that temporal information in order to better decoding. For that many approaches such as:

- RNNs are used to predict and use the temporal information hence extracting the temporal feature
- Gated Recurrent Neural Networks (GRUs) such as unidirectional and bidirectional GRUs are used previously.
- LSTMs are used to extract the time series information.
- Transformers are more recent approach which are being applied to extract temporal features
- In addition to all different mechanisms temporal convolutional networks (TCNs) which are another form of neural networks can be used to extract temporal features.

3.3.4 Combining Features and Classification

Starting with CNNs, these architectures have long been a key player to lip reading systems, specially in the frontend for spatial feature extraction. CNNs are good at capturing visual patterns and informations from lip movements, making them well-suited for the analysis of image data. However, 3D CNNs can also be used standalone model to decode temporal data

as well as spatial data. Transitioning to Transformer-based models, including attention mechanisms, we see a shift towards standalone methods for both spatial and temporal feature extraction. These models, such as Transformers, have gained popularity due to their ability to focus at the relevant portions of the input sequences. This capability allows them to capture complex spatial relationships and temporal dynamics effectively. However, despite the effectiveness of attention-based models, hybrid approaches have remained prominent in lip reading research. These architectures combine the strengths of CNNs for frontend spatial feature extraction with RNNs [5], such as LSTM [6] networks or GRUs [7], for backend memory units. CNNs excel at capturing spatial patterns from lip movements, while RNNs/LSTMs/-GRUs are very efficient at modeling temporal dependencies within sequential data, such as speech or lip movements. In essence, hybrid architectures leverage both spatial and temporal information effectively, resulting in remarkable accuracy and robustness in deciphering spoken language from visual cues. This collaborative approach underscores the continuous evolution and advancement in lip reading technology, promising ever-improving performance and understanding of visual speech data.

3.4 Our Proposed Method for Machine Lip-reading

For obvious advantages of hybrid models we have used a hybrid model consisting of 3D CNN and Bi-directional LSTM inspired from LipNet[42] . As data structure of LipBengal [43] was different than the Grid-corpus [44], original dataset used in LipNet [42] we had to work from scratch.

For that we have made several data loading, encoding and data dividig functions. Then to preprocess the data to detect face and facial landmarks we have used DL based method after that data is fed to the model and encoded detected output is decoded. And the steps are explained below:

3.4.1 Data in Context of Model

In the context of our model, the term "data" refers to a sequence of image frames and their corresponding alignments, as illustrated in the accompanying figure. The LipBengal dataset [43] consists of image frames depicting words in Bengali, alongside their respective alignments denoting the corresponding Bengali word for each frame sequence.

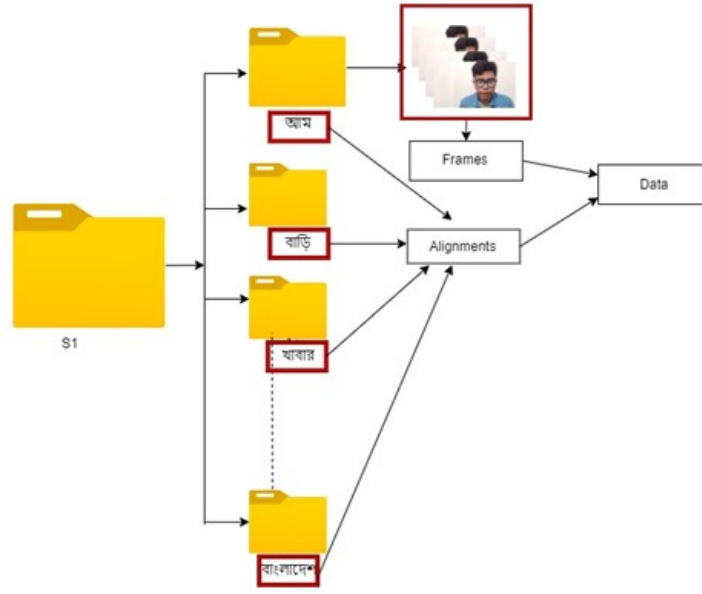


Figure 3.3: Data structure of the dataset:LipBengal

Data in general here contains two parts: one is the image sequences and other one is the word said mentioned as alignment here. From the data image sequences will go through several pre-processing steps and will be converted into 280 x 135 pixels in dimension. Each batch will have 45 frames. And alignments will be converted to numbers. All these will be fed as input to our model which will output a [45,75] tensor. Which will later be decoded into texts.

3.4.2 Data Loading Functions

To make our model learn well, we need to get the data from folders efficiently. For this, we've created some functions.

- **Load_word_images():**

This function goes through the word folders and grabs each image one by one. It uses a tool called Google MediaPipe to find faces and important points on them, especially the lips. Then, it focuses on the mouth area, makes sure the image looks right, and changes it to black and white. More on this in the preprocessing section.

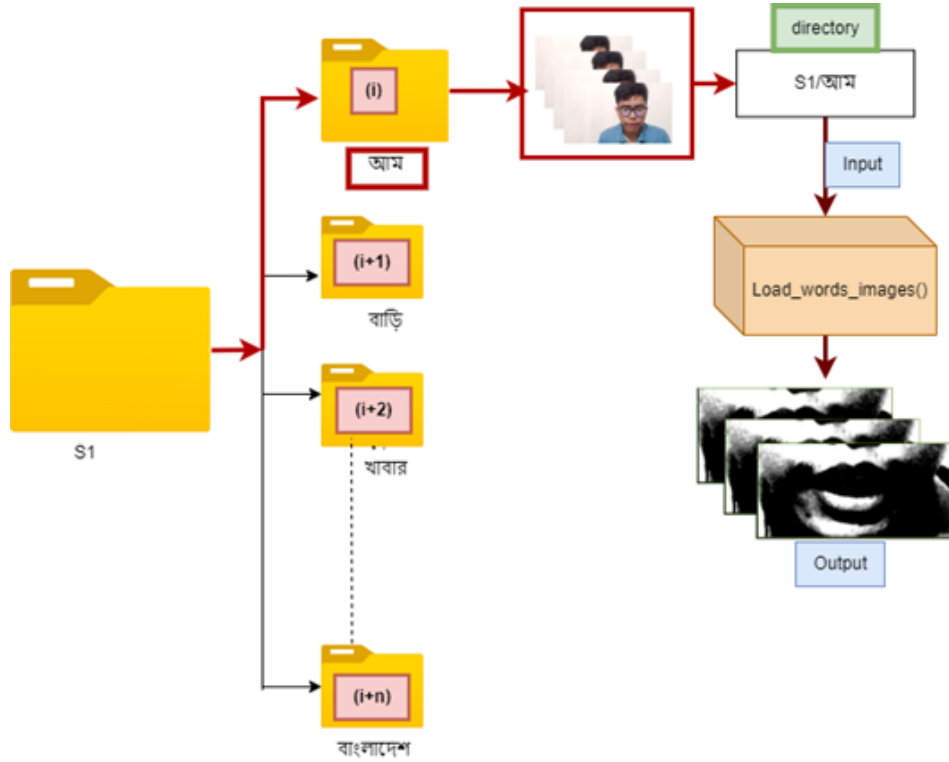


Figure 3.4: Working procedure of Load_word_images() function

- **Char_to_num() and num_to_char():**

These functions convert character data into numbers and numbers into characters. In our model we used 73 Bengali characters including alphabets, numbers and some special characters as classes.

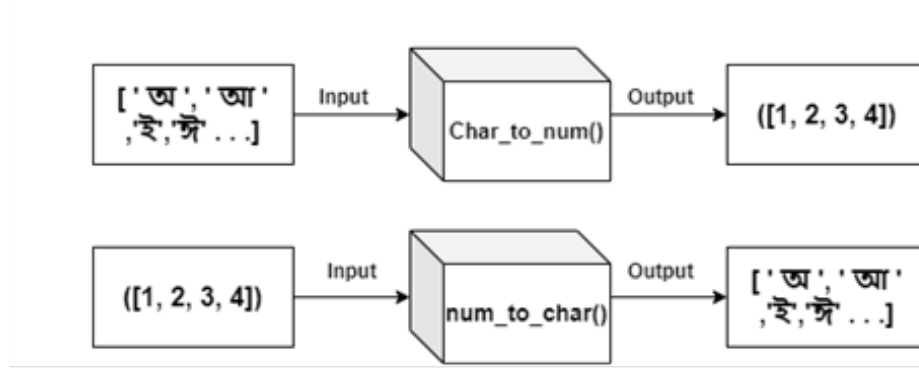


Figure 3.5: Working procedure of char_to_num() function

Character	number	Character	number	Character	number	Character	number
'অ'	1	'জ'	19	'ষ'	37	'৪'	55
'আ'	2	'ঝ'	20	'র'	38	'৫'	56
'ই'	3	'ঞ'	21	'ল'	39	'৬'	57
'ঈ'	4	'ট'	22	'শ'	40	'৭'	58
'উ'	5	'ঠ'	23	'ষ'	41	'৮'	59
'ঊ'	6	'ড'	24	'স'	42	'৯'	60
'ঋ'	7	'ঢ়'	25	'হ'	43	'০'	61
'এ'	8	'ণ'	26	'ড়'	44	'১'	62
'ঐ'	9	'ত'	27	'ঢ়'	45	'২'	63
'ও'	10	'থ'	28	'য়'	46	'৩'	64
'ঔ'	11	'দ'	29	'ৎ'	47	'৪'	65
'ক'	12	'ধ'	30	'ং'	48	'৫'	66
'খ'	13	'ন'	31	'ঃ'	49	'৬'	67
'গ'	14	'প'	32	'়'	50	'৭'	68
'ঘ'	15	'ফ'	33	'০'	51	'৮'	69
'ঙ'	16	'ব'	34	'১'	52	'৯'	70
'চ'	17	'ভ'	35	'২'	53	'০'	71
'ছ'	18	'ম'	36	'৩'	54	'১'	72
						'২'	73

Figure 3.6: Working procedure of num_to_char() function

- **Load alignments():**

This function looks through the folders inside the words folder. It saves the names of these folders into a list in Python. As machine learning model does not understand letters so we converted the list of characters to numbers using functions `char_to_num()` and later we convert back to numbers using `num_to_char()` function.

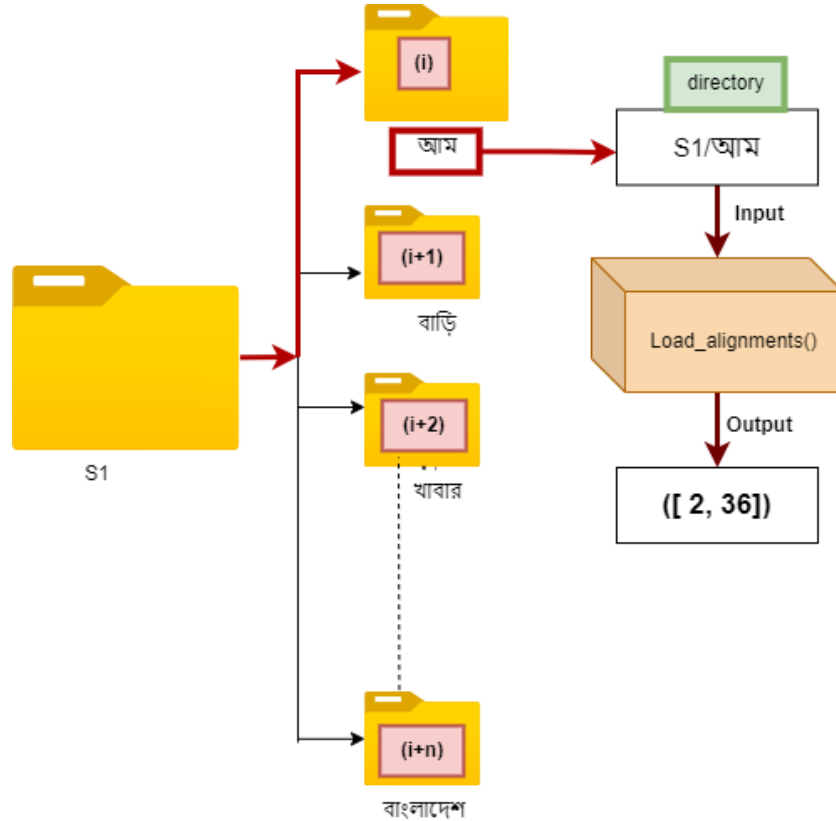


Figure 3.7: Working procedure of Load-alignment function

- **Load_data():**

This is the main function for getting the data. It puts together the work of the previous functions. It gets both the images and their alignments from the speaker's folders. This way, we make sure we have all the data we need for training our model.

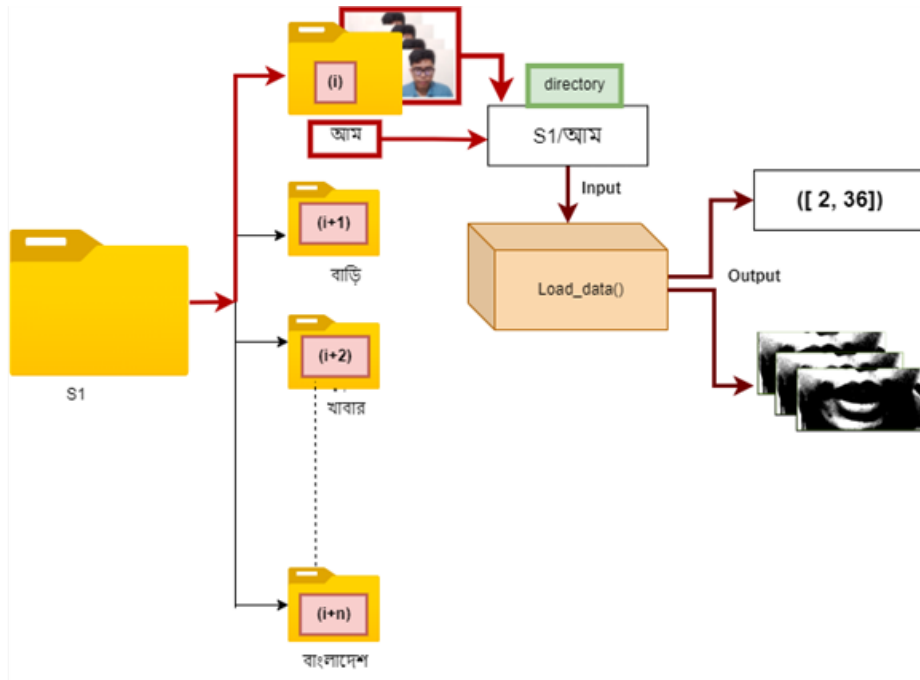


Figure 3.8: Working procedure of Load-data() function

These are the main operations that allow data to be loaded from the desired directory efficiently. These data loading algorithms can be applied to any dataset that is structured similarly to LipBengal.

3.4.3 Creating Data Pipeline

Data pipeline is a way to load data before feeding to a DL model. The way it works is all the data are loaded into a buffer then they are shuffled randomly and then it is divided into train and test set. 90% of the shuffled data are taken as train and the rest is taken as test set.

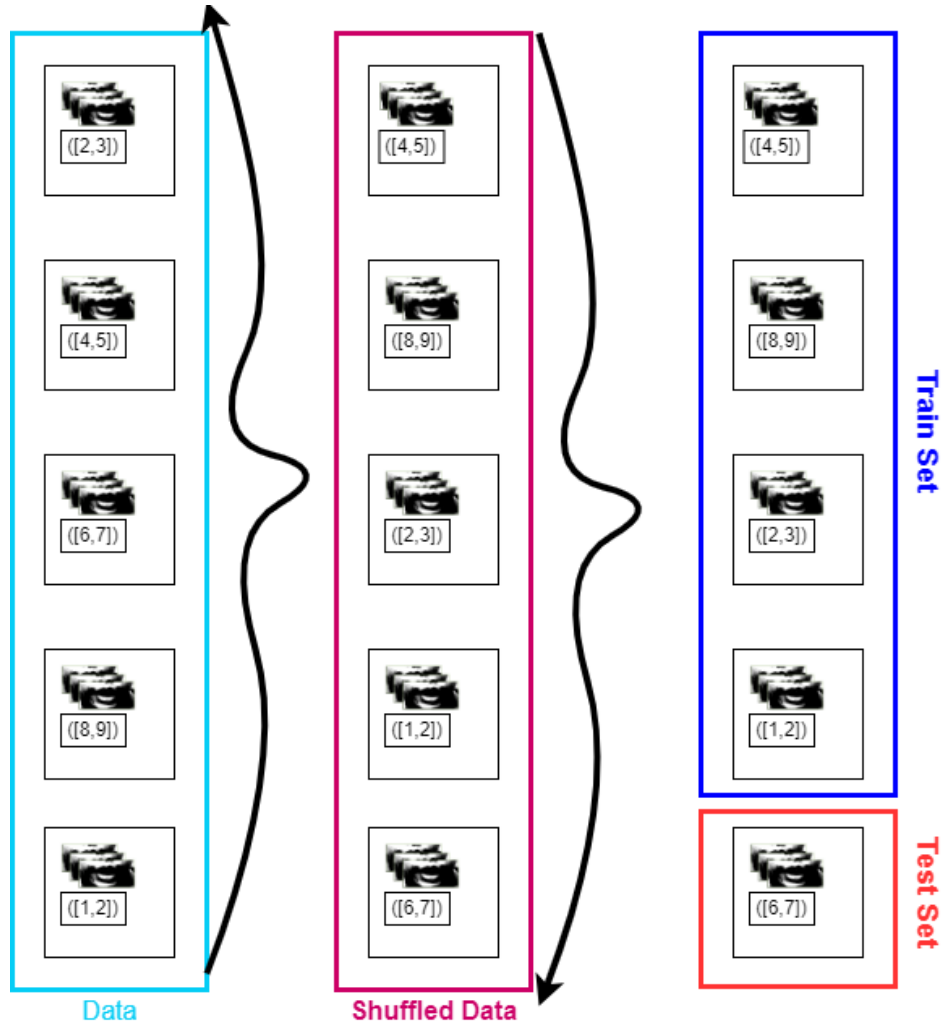


Figure 3.9: Steps involve in the creation of data pipelining

3.4.4 Image Data Preprocessing

Preprocessing is a crucial step in which we focus on isolating the ROI and reducing the dimensionality of the image within this area.

To isolate the ROI, We utilized Google MediaPipe Face Solutions to detect the speaker's face. From the 468 facial landmarks provided, we specifically targeted landmarks representing the upper lip (landmark[0]) and lower lip (landmark[17]). This targeted approach ensures that we capture the essential features necessary for subsequent processing steps.



Figure 3.10: Input image

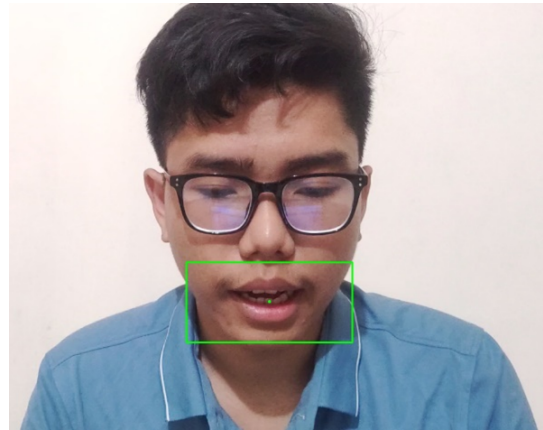


Figure 3.11: 468 Facial Landmarks

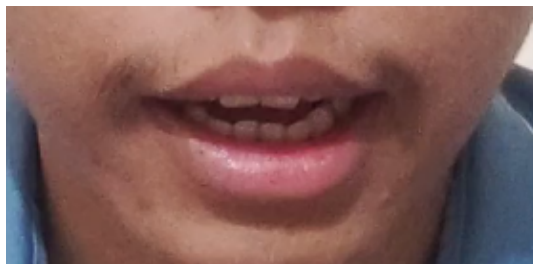
Once we identified the landmarks of the lips, we had to figure out their exact positions in terms of the image's height and width. Since the landmark coordinates were normalised, meaning they were scaled relative to the image size, we needed to convert them to actual pixel positions for accurate representation on the image. Once we obtained the actual coordinates, we proceeded to determine the center of the mouth by calculating the mean of the upper and lower lip coordinates shown in fig:a. Using this center point, we cropped the image to a size of 280x135 (width x height), ensuring the center of the mouth served as the focal point of the cropped image shown in figure 3.12(c).



(a) Center of mouth highlighted with green dots



(b) Lip localization



(c) Cropped image of dimension (280x135)

Figure 3.12: Cropping of lip-region

When a frame doesn't turn out just right, like in figure 3.13(a), we might get reduced dimension like Figure B. But if the picture isn't the size we want, like in 3.13(c), we pad the extra pixels with zero values. This makes sure all the pictures are the same size, like in figure 3.13(d).



Figure 3.13: Padding for non-ideal images

The padding is done to ensure all the images of a sequence have the same dimension as 280 X 135. Though our model does not require a fix dimension for input as it is coded intelligently to reshape its input as the largest dimension of an image of a batch. Yet to reduce complexity and for better analysis we needed to have fixed dimension of images.

3.4.4.1 Reducing Dimentionality

To reduce the dimensionality, we changed the RGB images to gray-scale, as seen in fig- 3.14(a). This means we only kept one channel of data instead of three. Then, to make sure all the images were on the same scale, we made some adjustments. We took the average value of all the pixels and subtracted it from each pixel's value. Then, we divided by the standard deviation to keep everything balanced, like in fig- 3.14(b).



(a) Grayscale image



(b) Standardized image

Figure 3.14: Procedure to reduce dimentionality

Images are further preprocessed while loading as batches to feed the DL model. If any sequence consists of less than 45 frames empty frames are padded. After that frames and alignments are loaded using data-pipeline.

3.5 Designing the DL Architecture

Our lip-reading model combines 3D CNNs for spatial features and Bi-directional LSTMs for temporal context. A dense layer with softmax activation classifies outputs. Training utilizes a custom Connectionist-Temporal-classification-Loss(CTC loss) function for backpropagation, enabling effective alignment of predicted phoneme sequences with ground truth labels. This integrated approach enhances the model's ability to decode visual speech cues, promising advancements in human-computer interaction and assistive technologies. A simple architecture can be visualized like below:

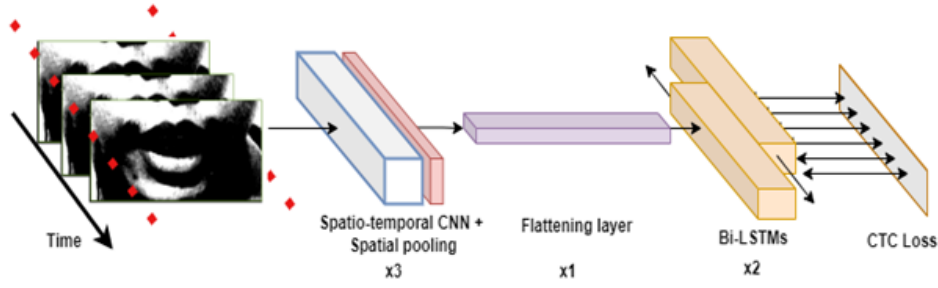


Figure 3.15: Workflow diagram of the Model

Here is more detailed architecture of the model:

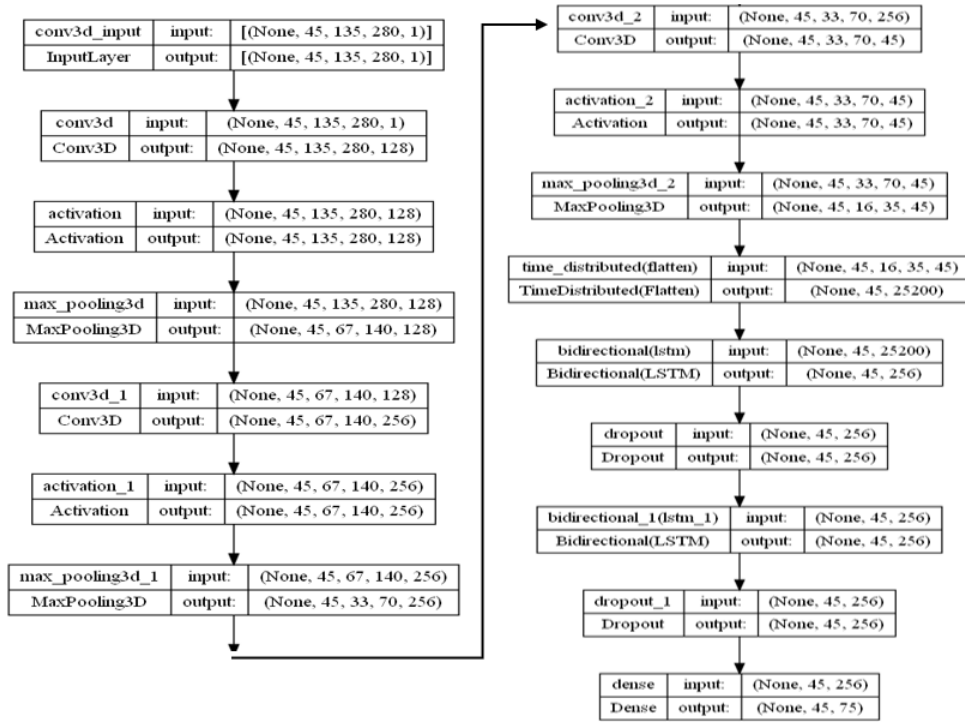


Figure 3.16: Detailed diagram of the model

3.5.1 Layers of the Model

Our model is made up of sub layers of 3D CNN and Bidirectional LSTMs. Each sub layer of 3D CNN consists of a CNN layer followed by a maxpooling layer and each LSTM layer is accompanied by dropout layer to prevent over-fitting. Input and output of each layer is described below:

- **Conv3D layers:** The first Conv3D layer with 128 filters and a kernel size of 3x3x3 processes the input data with shape (45, 135, 280, 1). With padding='same', the output shape remains the same as the input shape: (45, 135, 280, 128). Each filter produces a feature map, resulting in 128 feature maps. The subsequent Conv3D layers maintain the same output shape as the previous layer due to padding='same'.
- **MaxPool3D Layers:** Each MaxPool3D layer with a pooling window size of (1, 2, 2) reduces the spatial dimensions of the feature maps by half along the height and width axes, while keeping the depth (number of filters) unchanged. Therefore, the output shape after each MaxPool3D layer becomes (45, 67, 140, 128).
- **TimeDistributed Flatten Layer:** The TimeDistributed Flatten layer flattens the output from the convolutional layers across time, but it does not affect the spatial or depth dimensions. Therefore, the output shape remains $(45, 67, 140 * 128) = (45, 67, 17920)$.
- **Bi-directional LSTM Layers:** The Bidirectional LSTM layers process the sequential data bidirectionally, capturing temporal dependencies. As the input is a sequence of feature vectors with shape (45, 67, 17920), the LSTM units operate on each time step independently. The output shape of each Bidirectional LSTM layer is the same as the input shape, (45, 67, 256), where 256 is the number of units in each LSTM layer (128 units in each direction).
- **Dense Layer:** The final dense layer produces the output predictions. With units equal to the size of the vocabulary plus one, and using the softmax activation function, the output shape becomes (45, 67, vocabulary_size + 1). Each value in the output represents the probability of each class in the vocabulary.

To summarize the architecture our model takes input of 45 image frames each having dimension of 280 x 13 x 1 (width x height x channel). Which passes through bunch of 3D CNNs and max pooling layers after that these data is flattened to feed to bi-directional LSTMs and at last a fully connected layer to classify.

3.6 Loss Function and Optimizer Used in our Model

3.6.1 Loss Function

We have used 'CTC Loss' as a loss function. As in our case precise alignment between elements of the input sequence (image frames) and the corresponding output labels (alignments) is unknown. Traditional loss functions, like categorical cross-entropy (CCE), often require perfectly aligned data, rendering them unsuitable for these tasks. CTC addresses this issue by intelligently calculating probabilities over all possible alignments, allowing for flexibility and robustness in sequence-to-sequence models.

The key idea behind CTC is to introduce a 'blank' symbol along with the existing target symbols. It enables the model to predict repetitions of the same character, blanks (representing no output), or transitions between characters without relying on a pre-determined alignment. How CTC works can be broken down into some parts.

3.6.1.1 Formalization

Consider an input sequence $X = (x_1, x_2, \dots, x_T)$ and a target label sequence $Y = (y_1, y_2, \dots, y_U)$, where T and U represent the lengths of the input and output sequences respectively.

1. **Augmented Label Set:** CTC extends the label set with the blank symbol (-) to create a new label set L' .
2. **Possible Alignments:** All possible alignments between the input sequence X and the output sequence Y are generated. An alignment (represented as π) consists of elements from the augmented label set L' . The length of an alignment is always the length of the input sequence, and it's constructed by collapsing repetitions of the same character and removing blank symbols.

3. **Probability of an Alignment:** The probability of a particular alignment π is calculated as the product of individual symbol probabilities along the alignment:

$$P(\pi|X) = \prod_{t=1}^T y_{\pi_t}^t$$

where $y_{\pi_t}^t$ is the probability of emitting the symbol π_t at time step t , as predicted by the neural network.

4. **Total Probability of the Target:** To find the total probability of the target label sequence Y , CTC sums the probabilities of all possible alignments:

$$P(Y|X) = \sum_{\pi \in B(Y)} P(\pi|X)$$

where $B(Y)$ represents the set of all possible alignments that can be collapsed into the target sequence Y .

5. **CTC Loss Function:** The CTC loss is defined as the negative log-likelihood of the true label sequence. The goal of training is to minimize this loss:

$$L_{\text{CTC}} = -\ln P(Y|X)$$

3.6.1.2 Explanation of the Equation

1. **Augmented Label Set:** CTC introduces a "blank" symbol (-) to the standard label set. This enables the model to represent pauses, repetitions, or transitions between characters without a strict alignment.
2. **Alignments (π):** All possible ways to align the input sequence X with the target sequence Y are considered. An alignment, represented by ' π ', is constructed from the augmented label set and has the same length as the input sequence.
3. **Probability of an Alignment:** The probability of a single alignment is the product of the probabilities of each symbol along the alignment:

$$P(\pi|X) = \prod_{t=1}^T y_{\pi_t}^t$$

where $y_{\pi_t}^t$ is the probability of emitting the symbol π_t at time step t , as predicted by the neural network.

3.6.1.3 Computation

The efficient calculation of the total probability $P(Y=X)$ is achieved using a dynamic programming algorithm called the forward-backward algorithm. This involves:

- Forward Pass: Calculates the probability of all possible prefixes of alignments
- Backward Pass: Calculates the probability of all possible suffixes of alignments.

3.6.2 Optimizers

We have used Adaptive moment estimation (Adam) Optimizer for our task. A more detailed explanation is given below.

Adam is an efficient gradient-based optimisation algorithm designed to improve training speeds in deep neural networks. It leverages the strengths of past gradient information to adapt learning rates for each parameter, leading to faster and more stable convergence.

3.6.2.1 Formalization

1.Initialization:

- Initialise the first-order moment vector, m (mean of gradients), to 0.
- Initialise the second-order moment vector, v (squared gradients), to 0.
- Set hyperparameters:
 - β_1 : Exponential decay rate for the first-moment estimates (typical value: 0.9).
 - β_2 : Exponential decay rate for the second-moment estimates (typical value: 0.999).
 - ϵ : Small constant for numerical stability (typical value: 10^{-8}).

2. Gradient Calculation and Moment Updates For each iteration (t):

- Calculate the gradient g_t of the loss function with respect to the model's parameters.
- Update the first-order moment estimate: $m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$
- Update the second-order moment estimate: $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot (g_t)^2$

3. Bias Correction: Correct the bias in the moment estimates (especially during initial iterations)

- $m'_t = \frac{m_t}{1 - \beta_1^t}$
- $v'_t = \frac{v_t}{1 - \beta_2^t}$

4. Parameter Update Update the model's parameters θ using the corrected moments:

- $\theta_t = \theta_{t-1} - \alpha \cdot \left(\frac{m'_t}{\sqrt{v'_t + \epsilon}} \right)$
- α : Learning rate

3.6.2.2 Explanation of the Equations

- **Exponential Moving Averages (m, v):** Adam maintains exponentially decaying averages of past gradients (first moment) and squared gradients (second moment). These moving averages provide insights into the direction and scale of updates needed.
- **Bias Correction** Especially in early iterations, the moment estimates tend to be biased towards zero. The bias correction step addresses this, improving the estimates.
- **Adaptive Learning Rate** Adam adjusts learning rates for each parameter based on the first and second-order moments. Parameters with steeper gradients get smaller updates, while those with gentler gradients receive larger updates, aiding in faster convergence.

3.7 Training and Validation

As LipBengal is a large dataset , training and validation was done in batches where we took one speaker's data at a time, divided them in training a validation sets as explained in data pipeline section. We padded the image frames with empty frames to match 45 frames and padded the alignment with empty string to 20 characters. More detailed explanation is given below:

3.7.1 Training Procedure

- The training process involved iterating over the dataset in mini-batches, with each batch containing a variable number of image sequences and their corresponding word alignments
- During training, the model parameters were optimized to minimize the CTC loss using backpropagation.
- The training procedure was monitored using both training and validation datasets to evaluate the model's performance on unseen data.
- After each epoch, the model's performance metrics, such as loss and accuracy, were recorded for analysis.
- The training process continued for a predetermined number of epochs, with the model's parameters being updated iteratively to improve performance.

3.7.2 Callbacks

- Callbacks were used during training to monitor and control the training process effectively.
- The 'ModelCheckpoint' callback was utilized to save the model weights periodically based on specific criteria, such as minimizing the training loss.

- Additionally, the 'LearningRateSchedulercallback' was employed to adjust the learning rate dynamically during training

3.7.3 Evaluation Metrics

- The model's performance was evaluated using appropriate evaluation metrics, such as word error rate (WER) or character error rate (CER), to assess the accuracy of the predicted word sequences compared to the ground truth alignments.
- These metrics provided quantitative measures of the model's performance and were used to validate its effectiveness for the given task.

3.8 Chapter Summary

This chapter outlines the proposed method for machine lip-reading, including several essential operations and techniques. The chapter begins by addressing the transformation of characters into numbers and vice versa, which is crucial for handling character data in the model. It then presents functions for loading alignments and data, which are vital to preparing the dataset for training.

The chapter discusses the development of a data pipeline, detailing its function in loading and preprocessing data prior to inputting it into the DL model. The following section provides a detailed explanation of data preprocessing steps, such as isolating the ROI and standardising image dimensions. Facial landmark detection and cropping techniques are used to emphasise crucial features for lip-reading. The chapter also covers image data preprocessing, highlighting the significance of reducing dimensionality and maintaining uniform image sizes by using padding. Detailed explanations are given on methods for reducing dimensionality, such as converting RGB images to grayscale and standardising pixel values.

The chapter discusses the design of the DL architecture, which integrates 3D CNNs for spatial features and Bi-directional LSTMs for temporal context. Diagrams depict the architecture,

while detailed descriptions outline the functionality and input-output specifications of each layer. The chapter also covers the loss function and optimizer utilised in the model, highlighting the implementation of CTC loss for accurate alignment and the Adam optimizer for effective gradient-based optimisation. Equations and computation details for both are provided for clarity.

The text outlines training and validation procedures, such as mini-batch iteration and model monitoring, and discusses the use of callbacks for effective training control. Evaluation metrics like WER and CER are used to evaluate the performance of a model.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

Research on improving deep learning models for visual speech recognition (VSR) is still essential, especially when looking at languages with little resources like Bengali. Our main goal in this thesis was to advance this emerging subject by creating an end-to-end system designed especially for Bengali VSR using a brand-new dataset that was thoroughly prepped for this use. We believe that our method is groundbreaking because it is the first of its kind in a Bengali language setting. The implementation of a hybrid model, which was skillfully designed to capture each aspect of visual speech data present in the Bengali language, was crucial to our research. We used a rigorous approach in both the training and validation stages, dividing our dataset into separate testing and training sets and making sure that sample selection for validation was random to reduce bias. We conducted a thorough assessment of our model's effectiveness by comparing its performance to industry standards that have been set.

We present a thorough examination of several performance metrics, such as precision, word error rate (WER), character error rate (CER), loss, accuracy, and F1 score. These measures are important benchmarks that provide important information about how well our model captures the unique characteristics of Bengali VSR. This chapter establishes the foundation for future research and developments in this field by offering a solid structure and thorough assessment, which will promote ongoing gains in the field of VSR for less resourceful languages like Bengali.

4.2 Experimental Setup

The evaluation of a deep learning model may differ based on the evaluation configuration used. We utilised batch training approaches to train our deep learning model due to the substantial computer resources required by the large LipBengal dataset. We allocated the tasks among our fellow researchers to expedite the training process. We utilised a local workstation equipped with an NVIDIA-3060 GPU and Google Collaboratory for training our model. Checkpoints have been saved and restored at intervals during the training model before moving on to the next batch. We used batches to train our model, with each batch containing frames and alignments of words uttered by a single speaker. We utilised randomised training and test separation to ensure unbiased evaluation ratings.

We obtained the performance parameters referenced in chapter 4.5 by utilising our partially trained model on the dataset. We randomly selected a sample from the test section of the dataset and used our model to predict the alignment. We have assessed many performance metrics by comparing the predicted alignment with the actual one. We utilised scikit-learn metrics to assess performance in Python.

4.3 Comparative Performance Analysis of Lip-reading Models

Lip-reading, or VSR, is an active area of research. Throughout the year, many researchers studied and developed many models for the field. Their work and methods are depicted here through bar charts, and a brief analysis of their works is shown so that anyone who will work on this topic can get a head start by this thesis.

A performance analysis of lipreading models over different lip-reading benchmarking datasets is shown in figure 4.1. It is being analysed with the informations collected from [45], [46], [47], [48], [49], [50], [42], [9], [51], [10], [52], [11], [53], [8], [54], [55], [56], [57], [14], [58], [59], [16], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [17], [70], [71], [72], [73], [74]

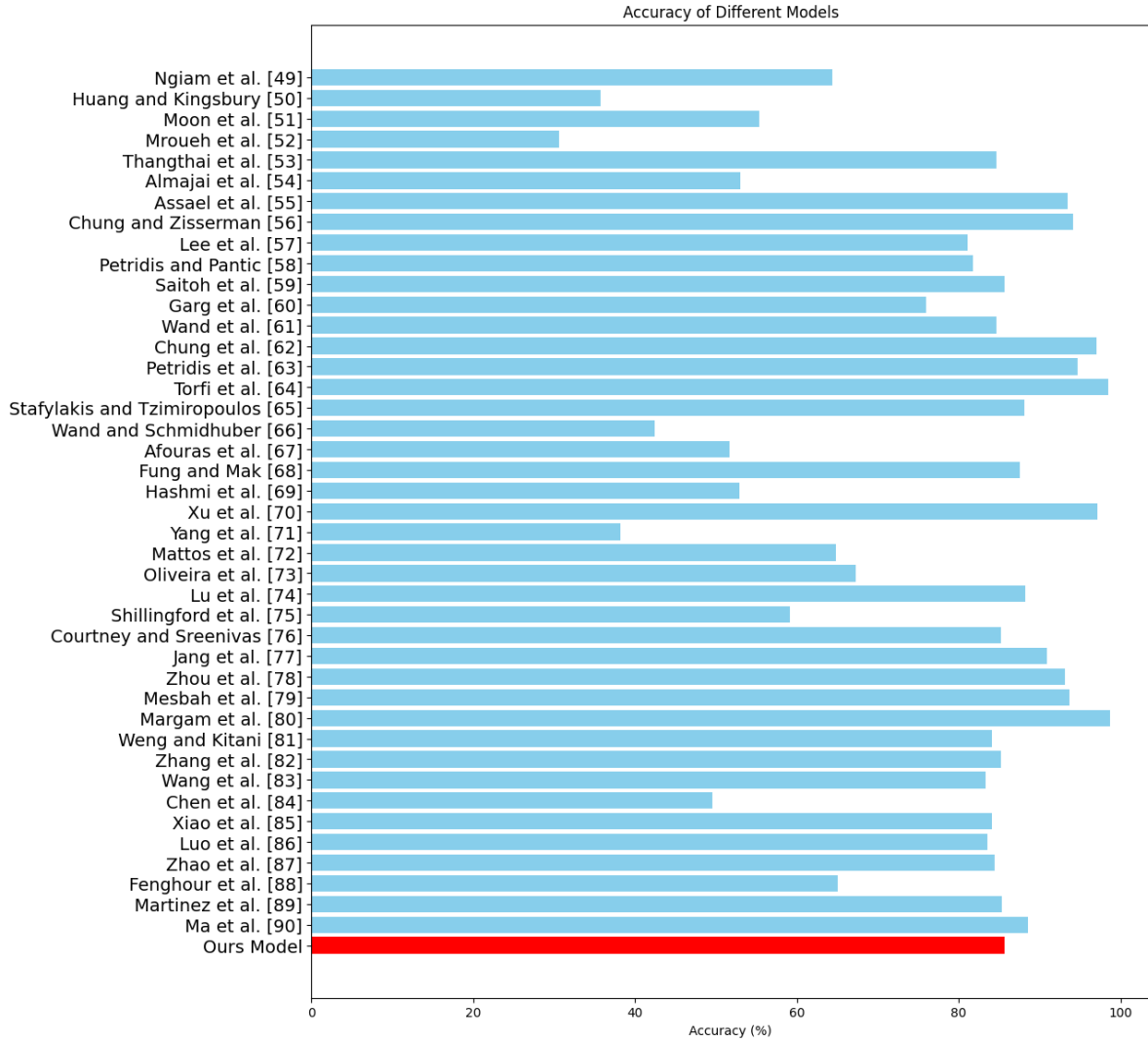


Figure 4.1: Accuracy of different Models

From the figure, we can see that the state of the art for lip reading with an accuracy of 98.7% is held by Margam et al. [69], who used a 2D+3D Convolutional Neural Networks (CNNs) for feature extraction and a Bidirectional Long Short Term Memory (LSTM) + Connectionist Temporal Loss (CTC loss) for classification. In our model, we trained up to a certain epoch and got an accuracy of about 84.61%, which is way above the average value.

Given that our model combines CNN and Recurrent Neural Networks (RNNs), we compared it to other structures of a similar design and discovered that [68] holds the top accuracy in this category with an accuracy of 87.5%.

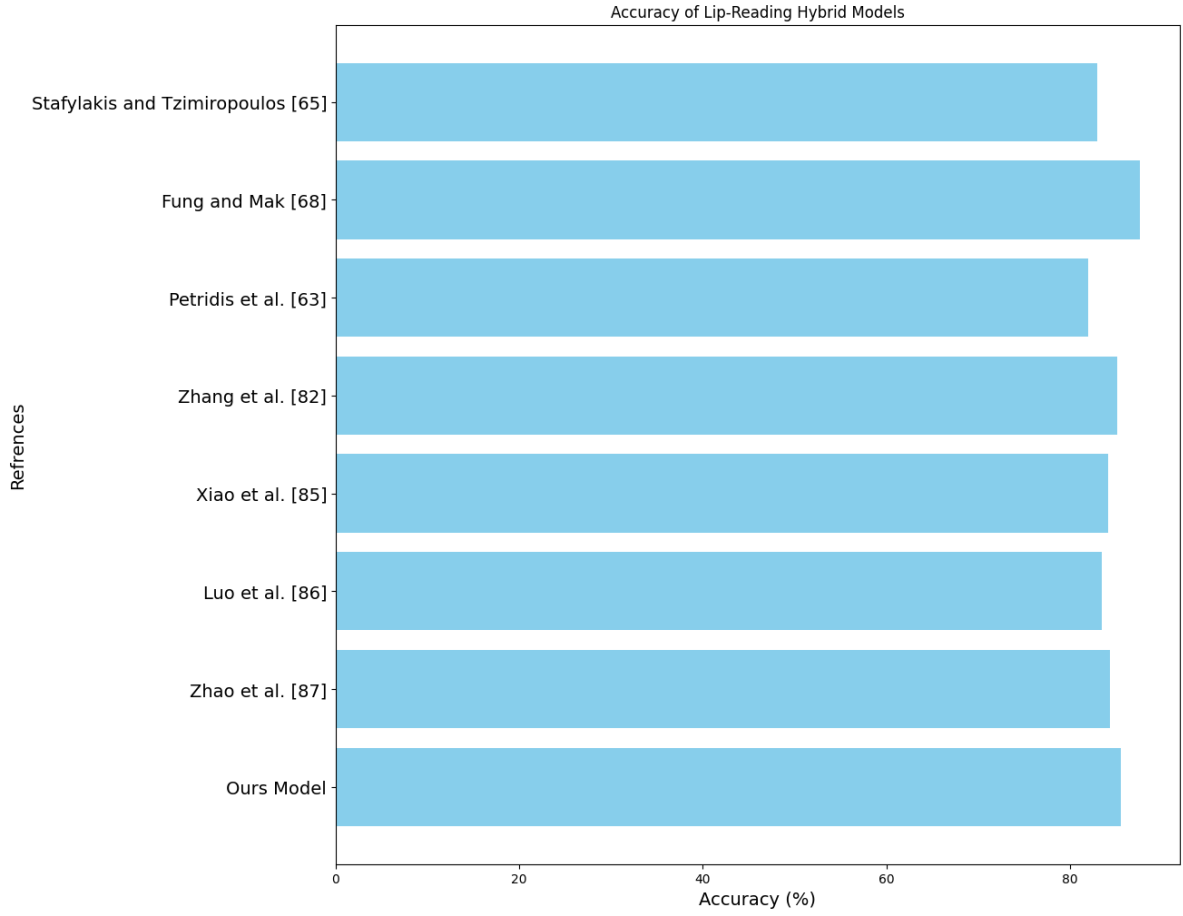


Figure 4.2: Accuracy of hybrid models

In this case, 3D CNN was used as a feature extractor, and a Bidirectional-LSTM model was used as a classifier. The classes were based on phrases, and the dataset used was ouluVS2.

As the classes of our model are Bengali alphabets, we compared the accuracy of our model with available phonemes and alphabet based class-segmented models. In In this category, our model holds the state of the art. The closest accuracy to ours is 84.67%, which was obtained by [53].

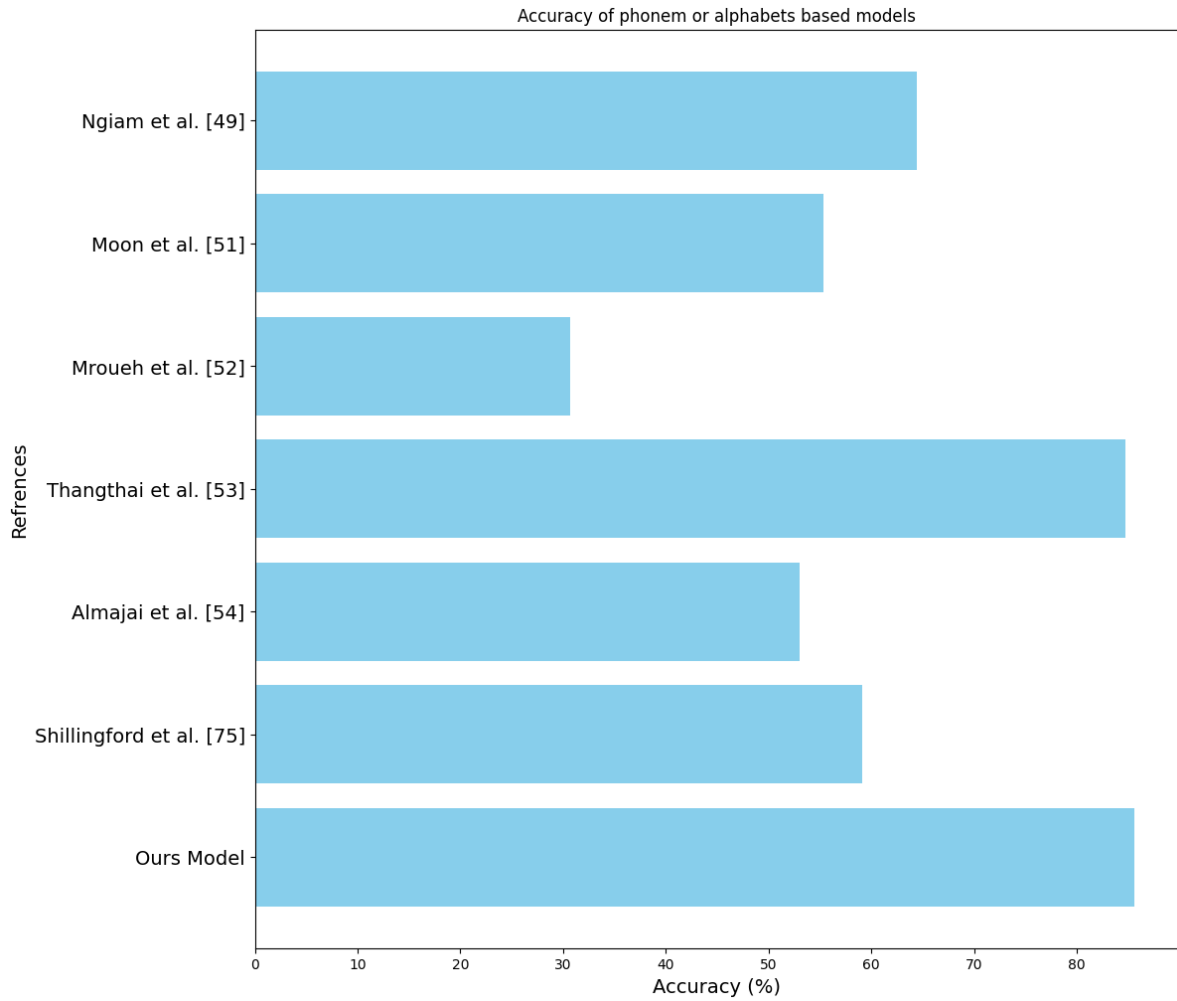


Figure 4.3: Accuracy of alphabet or phoneme based models

4.4 Analysis of Lipreading Models

Based on feature extraction and classification, we have plotted the performance of some models. For example, Fig. 4.4 shows that CNN-based feature extraction outmatches most of the feature extractors. A combination 2D+3D CNN worked best on the grid dataset, and [69] got about 98.7% accuracy.

We can also see the evolution of different feature extractors throughout time. From PCA to

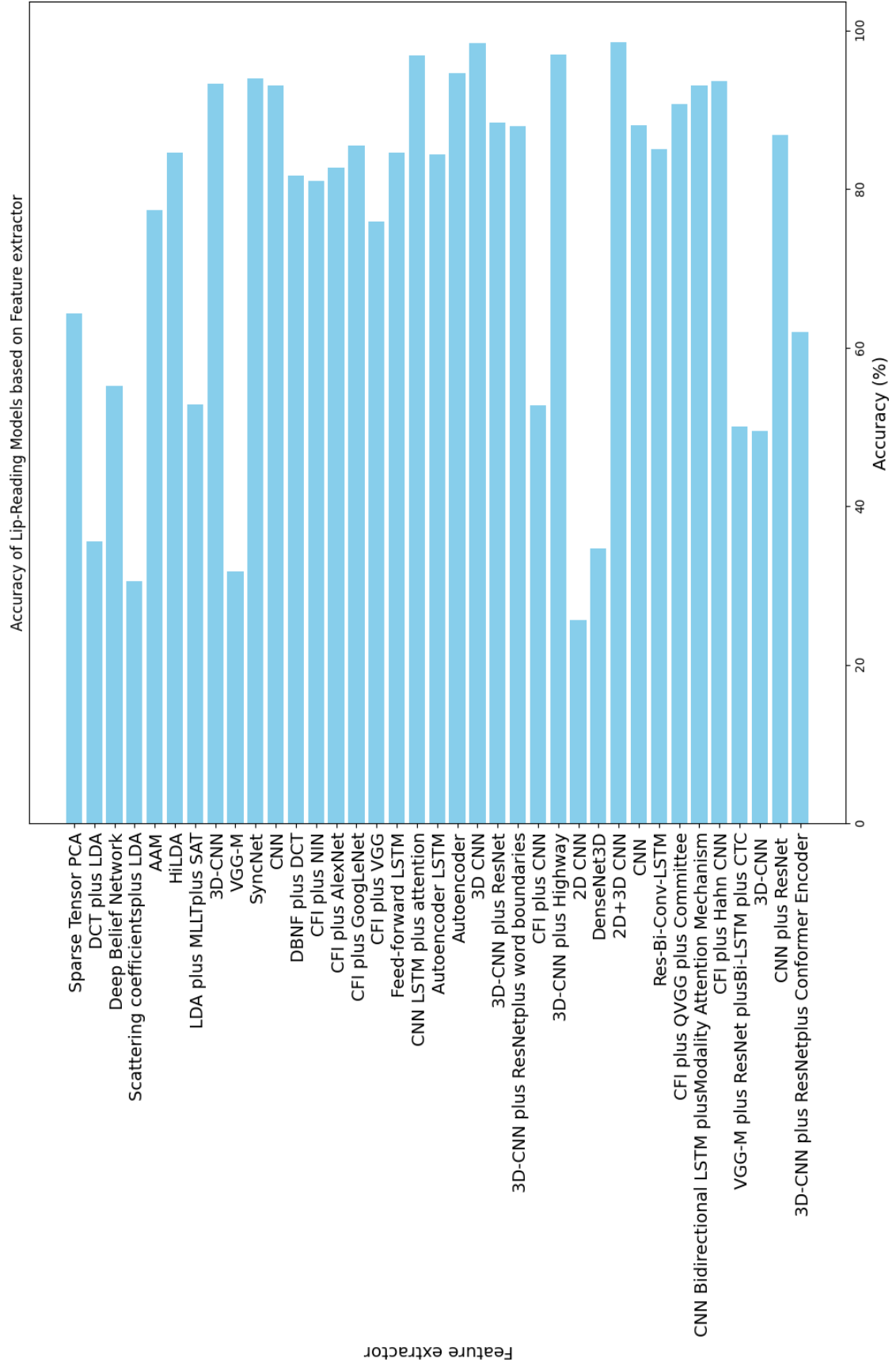


Figure 4.4: Performance based on feature extractor

3D CNN, the method to extract from images has evolved a lot. Still, CNNs holds the state of the art for feature extraction.

Based on the classifier, we have done the same analysis, as can be seen from the bar graph in Fig. 4.5.

Moreover, standalone or non-hybrid-type models are a way of solving this problem of lip-reading. The performance of non-hybrid models is shown in Fig. 4.6 below.

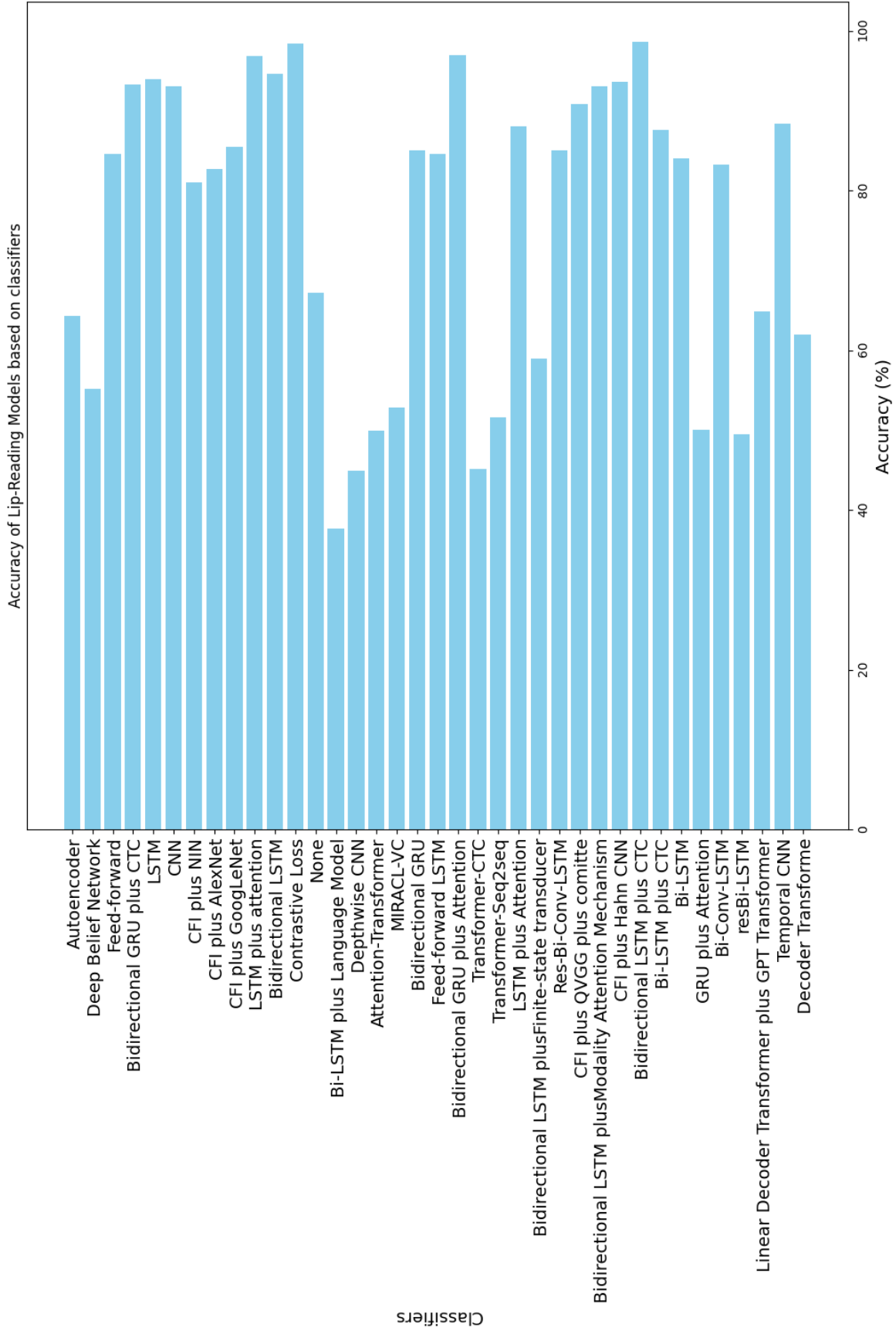


Figure 4.5: Performance based on classifier

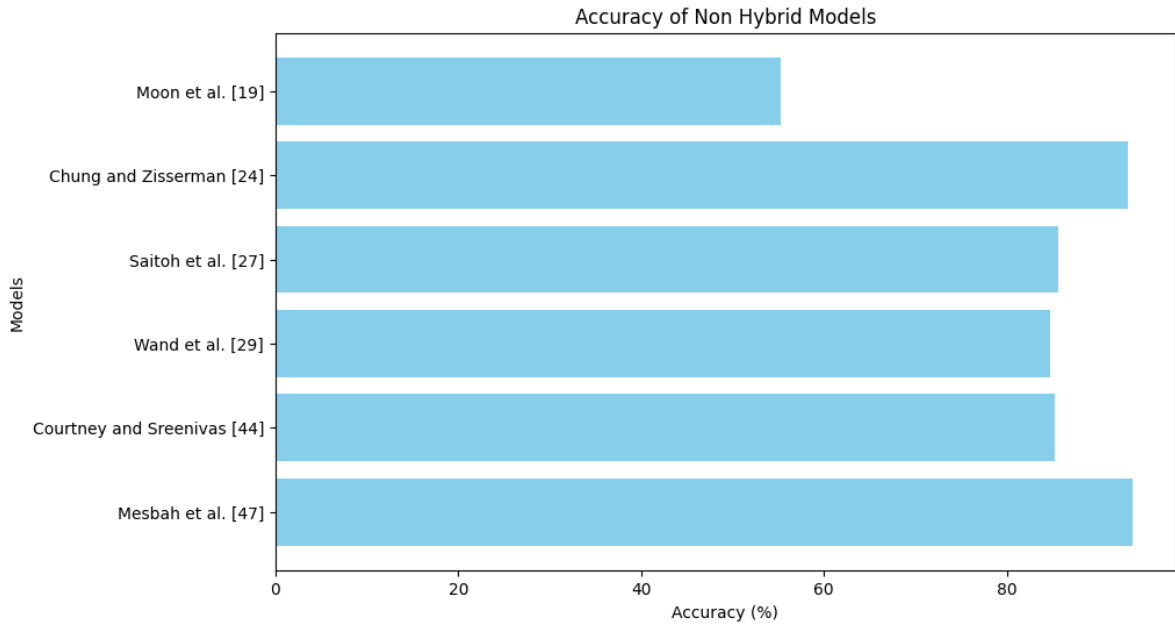


Figure 4.6: Performance of non-hybrid models

4.5 Evaluation Metrics

After running our model over LipBengal [2] dataset, the principal evaluation parameters we got are listed in Table 4.1

Table 4.1: Description of dataset:LipBengal

Name of the parameter	Value
Word Error Rate	0.34
Character Error Rate	0.087
Accuracy	0.846
Precision	0.8557
Recall	0.846
F1 Score	0.8504

4.6 Chapter Summary

This chapter of the thesis conducts an in-depth study of the performance of lip-reading models, with a specific emphasis on VSR within the Bengali language. The chapter starts with an introduction highlighting the significance of enhancing deep learning models for VSR in languages with inadequate resources, such as Bengali. The research aim is clearly defined: to create a comprehensive system designed for Bengali VSR using a newly collected dataset.

A comparative analysis is done on different lip-reading models to demonstrate the accuracy they achieve across various benchmark datasets. The thesis model performs significantly better than average, reaching 84.61% accuracy, and is identified as the most current model, achieving 98.% accuracy. Comparative analyses are performed using various model architectures, including hybrid models that combine CNN and RNN. The thesis model shows competitive accuracy in these analyses.

The chapter explores into the development of feature extraction methods and classifiers in lip-reading models, highlighting the perks of CNN-based feature extraction and the success of hybrid models in achieving high accuracy. Non-hybrid models are assessed to gain insights into various methods of lip-reading. The chapter provides a thorough assessment of the thesis model's performance on the LipBengal dataset, analysing metrics like WER, CER, accuracy, precision, recall, and F1 score. The metrics are important benchmarks for evaluating how well the model captures the distinctive features of Bengali VSR.

In this part of the thesis, we carefully looked at how well different lip-reading models work. We focused on showing how our model has improved and how it can help with understanding Bengali speech visually. This sets a good starting point for future research in this area.

CHAPTER 5

CONCLUSIONS AND FUTURE WORKS

5.1 Conclusions

In the ever-changing world of artificial intelligence, visual speech recognition (VSR), commonly referred to as lip-reading, presents a complex problem to be solved. As it is a very language-specific task, we have pioneered the foundation for Bengali lipreading through our thesis. We have used the LipBengal dataset to create an effective deep learning (DL) model that can assist eager individuals in solving similar challenges. We have carried out many procedures, from establishing an appropriate data loading method to identifying visual speech using the trained model. Earlier sections thoroughly explained these methods. During the journey, we encountered several challenges and successfully resolved multiple issues. We faced a major challenge because of the limited availability of datasets, prompting us to create our own. Working with Bengali was a significant issue due to the insufficient support for Bengali in many Python packages, necessitating multiple encoding and decoding processes. A challenge for new researchers in the field is that widely-used Python libraries like OpenCV do not have good support for Bengali, requiring alternative methods to process data. Navigating directories in Python when they contain Bengali characters is one of the main challenges. Establishing and configuring a local machine for DL could be a significant obstacle for new researchers. Training a large-scale DL model with limited resources can be very difficult. We handled this task using Google Colab, a local machine, and distributing the workload. We expect our work to provide comprehensive guidelines and a foundation for new scholars in this field. Our model will be the very first proposed model for Bengali lipreading until the date of publication. In conclusion, with the accuracy of 84.61% of our model, it can be stated that we have managed to "depict the whisper of Bengali."

5.2 Key Findings

- a. One of the key findings of our work will be the base model. The model was modified for Bengali in accordance with the LipNet architecture, which is considered state-of-the-art in this field.
- b. A more realistic preprocessing approach aimed at detecting facial landmarks on videos generated by mobile devices was utilized. This approach successfully detects and minimises the time- and cost-complexities of image frames.
- c. A fully trained model can easily be deployed on various embedded machines to recognise Bengali visually, thus aiding in visual or audiovisual communication.
- d. An analysis of previous lipreading models has been conducted, which can serve as a valuable resource for new researchers seeking additional information.
- e. Furthermore, machine lip-reading presents considerable challenges, and guidelines for this field, particularly for Bengali, are exceedingly limited. It is hoped that our work will serve as a guiding reference in this endeavour.

The application of this technology is constrained only by our imagination. Therefore, it is anticipated that the findings of this work will contribute to further developments in this and related fields.

5.3 Research Significance

Pioneering new technology has significance but comes with many uncertainties and obstacles. This work is expected to pioneer a new era in Bengali lipreading. When this research is applied in practical situations, significant enhancements are expected in cases where audio-based communication is challenging. The ability to understand conversations in video without audio is highly useful in different scenarios. The research clearly shows the importance of expanding lipreading technology and how it may improve communication in various situations.

5.4 Future Works

In the near future, the increasing presence of robots as companions will provide an ideal environment for implementing our research, especially in improving their visual perception. There exists a significant possibility for improvement by enhancing performance through hyperparameter tweaking and layer modifications, which can help new researchers achieve top-of-the-line standards.

Our work encourages the trend of using innovative ways and creating improved models in the ever-changing field of spatio-temporal problem-solving techniques. Future research should focus on creating vision transformer-based systems for lipreading issues, especially in the context of Bengali language, with a particular emphasis on word- or sentence-level analysis. We look for that the findings from our research will lead to fresh ideas and creative solutions among aspiring researchers.

Moreover, researchers have numerous possibilities to enhance their proficiency in Bengali language processing using Python, as these skills will become more valuable in developing lipreading technologies. We contend that our methods and study results serve as the foundation for future advancements, influencing the direction of innovative studies in this field.

5.5 Chapter Summary

The fifth chapter of the thesis discusses the significance and potential future paths of the research studied on Bengali lip-reading. The chapter highlights the pioneering nature of the work, emphasizing the challenges faced in dataset limited availability and language-specific processing. The key findings involve creating a foundational model based on the LipNet architecture, implementing effective preprocessing techniques for videos captured on mobile devices, and exploring the possible use of the trained model across different applications. The research is important because it contributes to the advancement of Bengali lip-reading, which helps in audiovisual communication when there is no audio present. Future attempts

involve implementing the developed technology in robotics and motivating new researchers to investigate hyperparameter tuning and innovative approaches to enhance lip-reading models. The emphasis will be on transformer-based solutions and language-specific obstacles. The chapter emphasises the significance of the research and its potential for future advancements in the field.

LIST OF PUBLICATIONS

Journal Papers:

- (i) **M. T. R. Sahed, M. T. Ahsan** and T. Aronno, R. Rafiul and H. Nyeem, *BLipNet: Towards Bengali Lipreading*, (With Supervisor)
- (ii) T. Aronno, R. Rafiul and **M. T. R. Sahed, M. T. Ahsan** and H. Nyeem, M. A. Wahed, M. K. Kundu and M. J. A. Sadeef, *LipBengal: Pioneering Bengali Lip-Reading Dataset for Pronunciation Mapping via Lip Gestures*, (With Supervisor).

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